



Coupled cluster and dislocation dynamics modeling of microstructure evolution in irradiated materials

Yang Li^a, Matthew Maron^b,^{*} Kristopher Baker^d, Benjamin Ramirez Flores^d, Thomas Black^e, James Hollenbeck^e, Inam Lalani^b, Nasr Ghoniem^c, Giacomo Po^b,^{*}

^a Shanghai Institute of Applied Mathematics and Mechanics, Shanghai Key Laboratory of Mechanics in Energy Engineering, School of Mechanics and Engineering Science, Shanghai University, Shanghai 200042, China

^b Department of Mechanical & Aerospace Engineering, University of Miami, Coral Gables, FL 33146, USA

^c Department of Mechanical and Aerospace Engineering, University of California Los Angeles, Los Angeles, CA 90095, USA

^d Knolls Atomic Power Laboratory, Fluor Marine Propulsion, LLC, Niskayuna, NY 12309, USA

^e Bettis Atomic Power Laboratory, Fluor Marine Propulsion, LLC, West Mifflin, PA 15122, USA

ARTICLE INFO

Keywords:

Cluster diffusion
Dislocation climb
Bias factors
Irradiation growth

ABSTRACT

We develop here a coupled cluster and dislocation dynamics framework to study the microstructure evolution of irradiated materials. The framework not only accounts for the three dimensional diffusion of radiation-generated clusters, but also their interaction with dislocation networks and the resultant climb motion of discrete dislocations within finite crystals. The framework is solved with a superposition solution scheme, and is applied to investigate the evolution of the irradiation-induced dislocation loops in zirconium (Zr), considering the effects of various bias factors including the diffusion anisotropy difference (DAD) of interstitials and interstitial clusters, the dislocation bias of defects to discrete dislocation segments, and the production bias of defects from the radiation cascade. We find that the DAD is the most critical factor influencing the kinetics of the loop evolution in Zr, while the recombination/interaction of mobile defects can induce a strong spatial dependence of the loop evolution together with the DAD. The method is also adopted to study the evolution of interstitial $\langle a \rangle$ and vacancy $\langle c \rangle$ dislocation loop ensembles consistent with the microstructure observed during irradiation-induced growth of Zr. Our findings not only reveal the spatial dependence of the size and ellipticity of the dislocation loops, but also suggest a limit on the anisotropy factor of interstitials to reproduce the co-growth of $\langle a \rangle$ and $\langle c \rangle$ loops in zirconium, in good agreement with experimental observations and other simulation results.

1. Introduction

Point defects in crystalline materials consist of two “opposite” types, vacancies and interstitials, which significantly influence their mechanical, electrical and thermal properties (Freyssoldt et al., 2014). At finite temperature, crystalline materials contain vacancies and self interstitial atoms (SIAs) with thermal equilibrium concentrations, depending on their formation energies (Hirth et al., 1983). In irradiated materials, incident high-energy particles, such as neutrons and ions, result in numerous displacements of atoms from their lattice sites, generating a large number of vacancies and SIAs in addition to those at thermal equilibrium concentrations (Was, 2007). These supersaturated defects evolve through recombination, annihilation at sinks, such as dislocations

^{*} Corresponding authors.

E-mail addresses: mlm335@miami.edu (M. Maron), gpo@miami.edu (G. Po).

<https://doi.org/10.1016/j.jmps.2025.106366>

Received 7 April 2025; Received in revised form 24 July 2025; Accepted 13 September 2025

Available online 23 September 2025

0022-5096/© 2025 Elsevier Ltd. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

and grain boundaries, and agglomeration into larger clusters such as dislocation loops and voids. Among these, the diffusion and interaction of defects with dislocations are of particular interest as they play a central role in many radiation-induced phenomena such as swelling, irradiation-enhanced creep, and irradiation-induced growth (IIG). Thus, the objective of this work was to develop a general framework to investigate the microstructure evolution of irradiated materials, which includes the diffusion and reaction of defect clusters, coupled with the climb motion of a discrete dislocation network.

Over the past decades, extensive efforts have been devoted to studying the evolution of defects, with no exception to modeling of the diffusion and interaction of defects with dislocations. As dislocations are moving sinks/sources for the defect population, coupling the diffusion of point defects with the dislocation motion is challenging. Earlier developments were analytical, using time-space variable transformations (Lothe, 1960; Balluffi and Seidman, 1965), while today many advanced computational techniques are employed to address this problem, such as the boundary integral method (Rungamornrat and Senjuntichai, 2009) and the phase field method (Geslin et al., 2014). In the context of discrete dislocation dynamics (DDD) simulations, Mordehai et al. (2008) analytically solved the dislocation climb velocity from a simplified boundary value problem for vacancy diffusion and modeled the climb of prismatic loops. In succession, Bakó et al. (2011) studied the vacancy mediated dislocation climb motion by incorporating an effective dislocation climb mobility law, which has been also adopted in many subsequent investigations (Danas and Deshpande, 2013; Liu et al., 2014; Ayas et al., 2014; Gao et al., 2019). In addition, other methods such as the second-kind integral equation formulation and the localized absorber solution have been developed in recent years to tackle creep (Jiang et al., 2017) and static recovery (Kohnert and Capolungo, 2022) of metals. Among these, a method on the basis of a Green's function formalism is of great interest since it can naturally capture the interaction between defects and climbing dislocations of arbitrary 3D geometry (Gu et al., 2015). This method has been widely used in investigating complex climb/diffusion problems, including the influence of surfaces, pipe diffusion, and boundary conditions (Rovelli et al., 2018; Breidi and Dudarev, 2022; Po et al., 2022). However, it should be noted that these works neglect the contribution of SIAs to the climb motion of dislocations. This is entirely reasonable for thermal cases, since the thermal concentrations of SIAs in metals are generally much lower than those of vacancies. However, under irradiation, a supersaturation of point defects and defect clusters is generated, strongly influencing the motion of dislocations. For instance, atomistic simulations have shown that radiation cascades near edge dislocations can promote the climb motion of dislocations (Fu et al., 2017; Derlet and Dudarev, 2020). By extending Mordehai's model, Thompson et al. (2018) have shown that the growth/shrinkage of a prismatic loop is dependent on the ratio of SIA to vacancy concentration. A model accounting for dislocation climb due to SIAs is thus widely desired.

Furthermore, significant attention has been paid to the calculation of bias factors to quantify the preferential absorption of the specific type of defects at sinks to understand the radiation damage in materials, e.g., swelling and creep (Brailsford and Bullough, 1972; Tomé et al., 1982). In the late 1950s, Greenwood et al. (1959) proposed a preferential absorption of SIAs at dislocations to explain void swelling, which is known as the "dislocation bias". The excess of interstitials arriving at dislocations also leads to their climb motion, giving rise to radiation-enhanced creep, and this concept has been applied to studying the long-term evolution of dislocation microstructures in irradiated materials (Wolfer, 1980, 2007). Later, more advanced molecular dynamics simulations of collision cascades revealed that SIAs are more likely to be formed in clusters within the cascade than their vacancy counterparts. This led Woo and Singh (1992) to propose a concept of "production bias", which avoids using unrealistic dislocation bias values. Furthermore, bias factors vary significantly with crystal structure. For noncubic metals, because of the crystallographic anisotropy, the diffusion rates of defects depend on the crystallographic directions. Consequently, a description of the microstructural evolution in non-cubic metals based on the isotropic diffusion concepts would be inadequate. In this circumstance, Woo (1988) suggested a bias concept of the diffusional anisotropy difference (DAD) to characterize the different absorption rates of interstitials and vacancies. This DAD concept has been used successfully to predict the IIG of zirconium, a common cladding material used in reactors (Choi et al., 2016; Li and Ghoniem, 2020; Ji et al., 2024). Although the concepts of bias are widely used in the study of microstructure evolution in irradiated materials, a selection of their values and an investigation of the coupling effect of different biases is still missing. This is particularly true for Zr, which is used as a reference material for this work, since the defect evolution involves the effects of dislocation bias, production bias, DAD, and self-induced preferential absorption (SIPA).

Due to irradiation damage and the aforementioned biases, Zr develops a complex microstructure, including mobile defects and clusters of interstitial and vacancy types, as well as prismatic (*a*-type) and basal (*c*-type) dislocation loops. The interactions between these defects lead to a dimensional change of Zr known as irradiation-induced growth (IIG) without applied loading, and irradiation-enhanced creep under applied loading. The long-term evolution of the Zr microstructure has been widely studied in the past by means of radiation damage rate theory models (Woo, 1988; Christien and Barbu, 2009; Barashev et al., 2015; Choi et al., 2016; Li et al., 2024b; Ji et al., 2024). By assuming a continuous representation of defect clusters and their interactions through deterministic equations, these models have provided significant insight into the effects of the alloying elements, sinks, and loop formation on the growth and creep of irradiated Zr. However, different models adopt different bias concepts to match to experiments. Indeed, Choi et al. (2016) suggests that the dislocation bias is the primary factor that results in anisotropic flux of interstitials towards sinks with different orientations, and Christien and Barbu (2009) shows that the DAD of SIAs plays a critical role in IIG. In contrast, Barashev et al. (2015) argues that the anisotropic flux of defects stems from the cascade production bias. A spatially-resolved cluster dynamics model including grain interaction stresses and free surfaces developed by Maron et al. (2024), also showed that the loop growth rate is not only dependent on the bias factors, but also spatial position within the finite volume. Unlike rate theory-based models that provide an averaged behavior description, object kinetic Monte Carlo (OKMC) models that account for the discrete and random nature of defect interactions have also been developed for studying the spatially dependent defect evolution in Zr (Christiaen et al., 2020; Sakaël et al., 2023). Since many methods adopt continuum field or point-like approximations, they ignore the elastic interaction between dislocations, which is considered a key mechanism at high dislocation density (Dai et al.,

2017; Breidi and Dudarev, 2022; Warwick et al., 2023). Additionally, these models assume that all loops are circular, even though the shape of the loop has been found to be an important factor that influences its reaction rate with other defects by theory and atomistic simulations (Saidi et al., 2020; Hulse and Race, 2021). TEM experiments of irradiated Zr have also observed that the type, shape and size of dislocation loops are intimately related with their spatial position, irradiation temperature and dose (Long et al., 2024; Li et al., 2025).

In this paper, we develop a general 3D DDD framework to investigate the coupled defect reaction–diffusion and dislocation climb in finite crystals subject to radiation conditions. The framework includes both vacancy and interstitial clusters, as well as different absorption biases to dislocations. We select zirconium as an example, where the emphasis is laid on unraveling the rate-limiting mechanism for dislocation loop evolution, and to study the IIG phenomenon. In Section 2, we present a general finite-deformation continuum formulation of discrete dislocation-based plasticity coupled to defect cluster diffusion. In Section 3, we show the discrete dislocation dynamics implementations, together with an effective description of the dislocation bias and diffusion anisotropy difference. In Section 4, the model is then validated by comparing the single-loop kinetics with theoretical predictions. A particular emphasis is then placed on the effects of diffusion anisotropy and mobile defect reactions. The effect of bias factors on the evolution of an individual and a population of loops is also systematically studied. A brief discussion on the results of the paper can be found in Section 5. Finally, the concluding remarks are discussed in Section 6.

2. The continuum framework of irradiated microstructure evolution

2.1. Kinematics of inelastic deformation

Let us consider the deformation of an infinitesimal *material* volume of monoatomic crystal containing a fixed number of atoms (fixed mass). The infinitesimal material element is labeled by the coordinate $\mathbf{X}$ in the Lagrangian (material) frame of reference, and by the coordinate $\mathbf{x} = \boldsymbol{\varphi}(\mathbf{X}, t)$ in the Eulerian (spatial) frame, where $\boldsymbol{\varphi}$ is the deformation map, and t denotes time. In the reference configuration the material is in its perfect crystal state with a volume of V_0 , and the atomic volume is Ω_0 . The deformation gradient tensor $\mathbf{F} := \partial \mathbf{x} / \partial \mathbf{X}$ of the element and its velocity gradient $\mathbf{L}$ can be decomposed into the inelastic component $\mathbf{F}^I$ and $\mathbf{L}^I$, as well as the elastic component $\mathbf{F}^E$ and $\mathbf{L}^E$, where the former process introduces defects in the perfect crystal and the latter one restores the overall compatibility (Bilby et al., 1957; Kröner, 1960; Lee, 1969). The two types of crystal defects considered in this work are **dislocations** and irradiation-induced **defect clusters**. Focusing on kinematics, the inelastic velocity gradient $\mathbf{L}^I$ is further decomposed into a dislocation-mediated plastic component $\mathbf{L}^P$ and defect-mediated compositional components $\mathbf{L}^D$ and $\mathbf{L}^V$. $\mathbf{L}^D$ is a velocity gradient in the flux of mobile defect clusters and $\mathbf{L}^V$ is associated with a volumetric deformation rate due to the relaxation of a defect cluster, which yields

$$\mathbf{L} = \mathbf{L}^E + \mathbf{L}^I = \mathbf{L}^E + \mathbf{L}^P + \mathbf{L}^D + \mathbf{L}^V, \quad (1)$$

Note that the current framework includes both point defects (interstitials and vacancies) and their clusters, as well as their interaction with each other and with dislocations, which is the major difference from previous work (Po et al., 2022). The objective is then to associate each of these deformation rates to its corresponding physical process. Now, considering the relative volume $J^I = dV^I / dV_0$, where dV^I is the volume of the configuration with the inelastic deformation, it can be expressed in terms of the defects concentrations as

$$\begin{aligned} J^I &= \frac{dV^I}{dV_0} = \det(\mathbf{F}^I) = \frac{\Omega_0 \# \text{atoms} + \sum_1^{Nv} \text{tr}(\chi_{0nv}^f) \# \text{cluster}_{nv} + \sum_1^{Ni} \text{tr}(\chi_{0ni}^f) \# \text{cluster}_{ni}}{\Omega_0 \# \text{atoms}} \\ &= \frac{\Omega_0 \# \text{atoms} + \sum_1^{Nv} (n\Omega_0 + \text{tr}(\chi_{0nv}^r)) \# \text{cluster}_{nv} + \sum_1^{Ni} (\text{tr}(\chi_{0ni}^r) - n\Omega_0) \# \text{cluster}_{ni}}{\Omega_0 \# \text{atoms}} \\ &= 1 + \sum_1^{Nv} \left(1 + \frac{\text{tr}(\chi_{0nv}^r)}{n\Omega_0} \right) \frac{n \# \text{cluster}_{nv}}{\# \text{atoms}} - \sum_1^{Ni} \left(1 - \frac{\text{tr}(\chi_{0ni}^r)}{n\Omega_0} \right) \frac{n \# \text{cluster}_{ni}}{\# \text{atoms}} \\ &= 1 + \sum_1^{Nv} \left(1 + \frac{\text{tr}(\chi_{0nv}^r)}{n\Omega_0} \right) c_{nv} - \sum_1^{Ni} \left(1 - \frac{\text{tr}(\chi_{0ni}^r)}{n\Omega_0} \right) c_{ni}. \end{aligned} \quad (2)$$

In the above equation, we define the concentration of vacancies or interstitials in the size- n cluster as $c_{n\star} := n \# \text{cluster}_{n\star} / \# \text{atoms}$. We also define $\text{tr}(\chi_{0n\star}^f) = \text{tr}(\chi_{0n\star}^r) \pm n\Omega_0$ where the + and - signs indicate vacancy or interstitial clusters, respectively. $\chi_{0n\star}^f$ and $\chi_{0n\star}^r$ represent the formation and relaxation volumes of n -size clusters (Goyal et al., 2019).¹

According to Eq. (2), the rate of J^I yields

$$\dot{J}^I = J^I \text{tr}(\tilde{\mathbf{L}}^I) = J^I \text{tr}(\mathbf{L}^I) = J^I [\text{tr}(\mathbf{L}^P) + \text{tr}(\mathbf{L}^D) + \text{tr}(\mathbf{L}^V)] = \sum_1^{Nv} \left(1 + \frac{\text{tr}(\chi_{0nv}^r)}{n\Omega_0} \right) \dot{c}_{nv} - \sum_1^{Ni} \left(1 - \frac{\text{tr}(\chi_{0ni}^r)}{n\Omega_0} \right) \dot{c}_{ni}. \quad (3)$$

where $\tilde{\mathbf{L}}^I = \dot{\mathbf{F}}^I \mathbf{F}^{-I}$ is the inelastic velocity gradient.

¹ If only vacancies or frenkel pairs accumulate in the material volume, and without considering effects from relaxation, then $\text{tr}(\chi_{0ni}^r) = n\Omega_0$ and $\text{tr}(\chi_{0nv}^r) = 0$, such that the volume change can be solely expressed by the concentration of vacancies as $J^I = 1 + c_{nv}$.

In addition, bearing in the mind that the number of atoms (mass) is conserved for a material volume, the defect concentrations thus obeys the balance law

$$\frac{\rho}{m_a} \dot{c}_{nv} = \frac{1}{J\Omega_0} \dot{c}_{nv} = -\nabla \cdot \mathbf{J}_{nv} + p_{nv} + F_{nv} \quad (4)$$

$$\frac{\rho}{m_a} \dot{c}_{ni} = \frac{1}{J\Omega_0} \dot{c}_{ni} = -\nabla \cdot \mathbf{J}_{ni} + p_{ni} + F_{ni} \quad (5)$$

where ρ is the spatial mass density, m_a is the atomic mass, $J = \det(F)$, while $\mathbf{J}_\star$ is the spatial defect flux and $p_\star$ is the production rate of defect clusters from dislocations. Other changes in defect concentration by the generation of the collision cascade and by the interaction of defect clusters, e.g., the recombination of vacancy and interstitial clusters, are all included in $F_\star$. Actually, in the general *cluster dynamics* framework, $F_\star$ also contains the reaction terms between species of defects. Neglecting higher order reactions, it can be written as

$$F_\star = \sum_{m \neq \star, n \neq \star} [R_2(m + n \rightarrow \star) c_m c_n - R_2(m + \star \rightarrow n) c_m c_\star + R_1(m \rightarrow \star) c_m - R_1(\star \rightarrow m) c_\star] + G_\star, \quad (6)$$

where c_m is the concentration of a generic defect species, with an index m that represents the number of interstitial atoms or vacancies contained in a cluster. The first and second terms of Eq. (6) correspond to second-order reactions involving two reacting species m and n , leading to the removal and generation of species $\star$, respectively. The third and fourth terms represent first-order reactions of conversion to and from species $\star$, respectively. The R_2 term describes the interaction rates between different pairs of mobile species, e.g. the recombination of single interstitials and single vacancies, $-\alpha_{iv} c_i c_v$ with α_{iv} being the recombination rate, while the R_1 term encodes the reactions that involve a single mobile species excluding its annihilation at dislocations, e.g. dissociation of di-interstitials, $-\alpha_{2i} c_{2i}$ with α_{2i} being the dissociation rate. Details about the cluster dynamics equations as well as reaction rates can be found in our previous works (Li and Ghoniem, 2020; Maron et al., 2024). It is important to note that the reactions between mobile defects and dislocations are discretely solved in dislocation dynamics framework (see $p_\star$ terms in Eqs. (4)-(5)), which is the main difference from the cluster dynamics method. The last term $G_\star$ represents the production rate of the defects from the cascade. It is worth noting that the conservation of vacancy and interstitial productions yields

$$\sum_1^{Nv} F_{nv} - \sum_1^{Ni} F_{ni} = 0. \quad (7)$$

Since the atomic volume in the stressed crystal is $\Omega = J^E \Omega_0$, from Eqs. (3) to (5) we have

$$\text{tr}(\mathbf{L}^I) = \Omega \left(\sum_1^{Nv} p_{nv} - \sum_1^{Ni} p_{ni} \right) - \Omega \left(\sum_1^{Nv} \nabla \cdot \mathbf{J}_{nv} - \sum_1^{Ni} \nabla \cdot \mathbf{J}_{ni} \right) + \sum_1^{Nv} \frac{\text{tr}(\chi_{0nv}^r)}{nJ^I \Omega_0} \dot{c}_{nv} + \sum_1^{Ni} \frac{\text{tr}(\chi_{0ni}^r)}{nJ^I \Omega_0} \dot{c}_{ni}. \quad (8)$$

In this work, Eq. (8) is the fundamental kinematic relation that allows us to associate the rates of deformation $\mathbf{L}^P$, $\mathbf{L}^D$, and $\mathbf{L}^V$ to the corresponding physical processes. A depiction and description of each physical process can be followed by Fig. 1. First, since the terms $p_\star$'s are the sources related to the dislocation climb process, then

$$\text{tr}(\mathbf{L}^P) = \Omega \left(\sum_1^{Nv} p_{nv} - \sum_1^{Ni} p_{ni} \right). \quad (9)$$

It should be noted that the interstitials and vacancies exert opposite contributions to the dislocation motion. Second, the fact that F^V is the volumetric deformation induced by the defect relaxation suggests that

$$\mathbf{L}^V = \sum_1^{Ni} \frac{\chi_{0ni}^r \dot{c}_{ni}}{nJ^I \Omega_0} + \sum_1^{Nv} \frac{\chi_{0nv}^r \dot{c}_{nv}}{nJ^I \Omega_0} = \sum_1^{Ni} \frac{\chi_{ni}^r \dot{c}_{ni}}{nJ \Omega_0} + \sum_1^{Nv} \frac{\chi_{nv}^r \dot{c}_{nv}}{nJ \Omega_0}, \quad (10)$$

where the volumes in the stressed crystals are $\Omega_\star = J^E \Omega_{0\star}$. Finally, we are left with the condition $\text{tr}(\mathbf{L}^D) = \Omega \left(\sum_1^{Ni} \nabla \cdot \mathbf{J}_{ni} - \sum_1^{Nv} \nabla \cdot \mathbf{J}_{nv} \right)$, which is satisfied by the following definition of the defect diffusion velocity “gradient”:

$$\mathbf{L}^D = \Omega \left(\sum_1^{Ni} \nabla \mathbf{J}_{ni} - \sum_1^{Nv} \nabla \mathbf{J}_{nv} \right). \quad (11)$$

Note that $\mathbf{L}^D$ describes the process in which the material volume changes shape due to a gradient in the flux of vacancies or interstitials. In the next section, we develop the irreversible thermodynamics of the defects microstructure evolution process to solve kinetic Eqs. (9), (10), and (11).

2.2. Irreversible thermodynamics

We now turn our attention to the irreversible thermodynamics of defect evolution. For simplicity, we consider a body under uniform temperature and irradiation. Let $\mathcal{V}_0$ be an arbitrary material volume, and $\mathcal{V} = \phi(\mathcal{V}_0)$ the corresponding spatial volume.

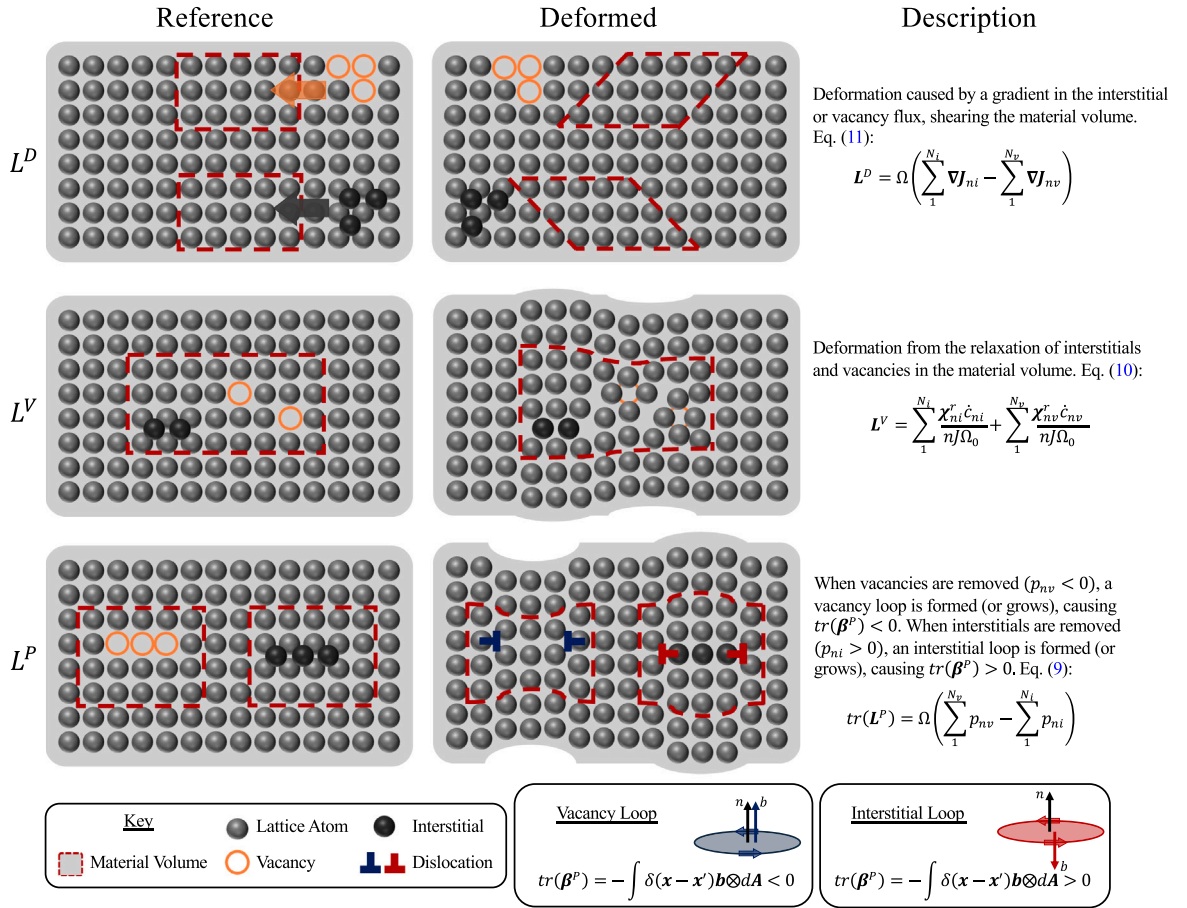


Fig. 1. Illustration of the kinematic framework. Depicted is a physical description of the various processes contributing to the inelastic deformation L^I , including L^P , L^V and L^D , along with a description and the corresponding equation from Section 2.1.

Mass is conserved within the material volume and the Clausius–Planck inequality has the standard form (e.g. [Malvern, 1977](#))

$$\int_{\mathcal{V}} d \, dV = \int_{\partial \mathcal{V}} \left(\mathbf{t} \cdot \mathbf{v} - \sum_1^{N_v} g_{nv} \mathbf{J}_{nv} \cdot \hat{\mathbf{n}} - \sum_1^{N_i} g_{ni} \mathbf{J}_{ni} \cdot \hat{\mathbf{n}} \right) dA - \int_{\mathcal{V}} \left[(\mathbf{v} \cdot \dot{\mathbf{v}} + \dot{\Psi} - \mathbf{f} \cdot \mathbf{v}) \rho - \sum_1^{N_i} \phi_{ni} F_{ni} - \sum_1^{N_v} \phi_{nv} F_{nv} \right] dV \geq 0 \quad (12)$$

where d is the internal dissipation density, $\mathbf{J}_\star$ the flux of defect clusters, $\mathbf{v}$ the material velocity, Ψ the Helmholtz free energy per unit mass, $\mathbf{f}$ the body force, and ρ the spatial mass density. $\phi_\star$ is the corresponding energy density injected by the external source. Note that the traction $\mathbf{t}$ and the Gibbs free energy per defect cluster $g_\star$ are quantities prescribed on the boundary $\partial \mathcal{V}$.

By introducing the Helmholtz free energy per vacancy or interstitial in a n -size defect cluster $f_\star$ and using the Gauss–Green theorem, the Clausius–Planck inequality can be written as ([Po et al., 2022](#))

$$\begin{aligned} \int_{\mathcal{V}} d \, dV = & \int_{\partial \mathcal{V}} \left[(\mathbf{t} - \boldsymbol{\sigma} \hat{\mathbf{n}}) \cdot \mathbf{v} - \sum_1^{N_v} g_{nv} \mathbf{J}_{nv} \cdot \hat{\mathbf{n}} - \sum_1^{N_i} g_{ni} \mathbf{J}_{ni} \cdot \hat{\mathbf{n}} \right] dA \\ & - \int_{\mathcal{V}} \left[\mathbf{v} \cdot (\rho \dot{\mathbf{v}} - \boldsymbol{\sigma} \cdot \nabla - \rho \mathbf{f}) - \boldsymbol{\Sigma} : \mathbf{L}^I + \sum_1^{N_v} f_{nv} \frac{\dot{\epsilon}_{nv}}{J \Omega_0} + \sum_1^{N_i} f_{ni} \frac{\dot{\epsilon}_{ni}}{J \Omega_0} \right] dV \geq 0. \end{aligned} \quad (13)$$

where $\boldsymbol{\sigma}$ is the symmetric Cauchy stress tensor, and $\boldsymbol{\Sigma} := \boldsymbol{\sigma} - W/J^E \mathbf{I}$ is the Eshelby stress tensor with W being the Helmholtz free energy density per unit intermediate volume, which is the stress measure conjugate to the inelastic velocity gradient $\mathbf{L}^I$. It is worth noting that the last terms associated with the defect production vanish due to the conservation of vacancies and interstitials.

This inequality is now used to derive relations between the internal irreversible rates of the deformation process and the corresponding forces. The independent fluxes shown in Eq. (1) are (a) the material velocity $\mathbf{v}$ associated with the tensor $\mathbf{L}$, (b) the defect fluxes $\mathbf{J}_\star$ defining the tensor $\mathbf{L}^D$, and (c) the dislocation motion related with the tensor $\mathbf{L}^P$. In the absence of viscosity, the dissipation d does not depend on the material velocity $\mathbf{v}$, yielding the standard balance of linear momentum in $\mathcal{V}$ and the

stress-traction relation on $\partial\mathcal{V}$:

$$\rho\dot{\mathbf{v}} = \boldsymbol{\sigma} \cdot \boldsymbol{\nabla} + \rho\mathbf{f} \quad \text{in } \mathcal{V}, \quad (14)$$

$$\boldsymbol{\sigma}\hat{\mathbf{n}} = \mathbf{t} \quad \text{on } \partial\mathcal{V}. \quad (15)$$

After substituting (9), (10), and (11) in (13), and by introducing the chemical potential per vacancy or interstitial in a n -size cluster, i.e., $\mu_{nv} := f_{nv} + \boldsymbol{\Sigma} : \boldsymbol{\chi}_{nv}'$, $\mu_{ni} := f_{ni} - \boldsymbol{\Sigma} : \boldsymbol{\chi}_{ni}'$, the Clausius-Planck inequality becomes

$$\begin{aligned} \int_{\mathcal{V}} d \, dV &= \int_{\partial\mathcal{V}} \left\{ \sum_1^{Nv} \mathbf{J}_{nv} \cdot [(\mu_{nv} - g_{nv}) \mathbf{I} - \Omega \boldsymbol{\Sigma}] + \sum_1^{Ni} \mathbf{J}_{ni} \cdot [(\mu_{ni} - g_{ni}) \mathbf{I} + \Omega \boldsymbol{\Sigma}] \right\} \hat{\mathbf{n}} \, dA \\ &+ \int_{\mathcal{V}} \left[\boldsymbol{\Sigma} : \mathbf{L}^P - \sum_1^{Nv} \mu_{nv} p_{nv} - \sum_1^{Ni} \mu_{ni} p_{ni} - \sum_1^{Nv} (\boldsymbol{\nabla} \mu_{nv} - \Omega \boldsymbol{\Sigma} \cdot \boldsymbol{\nabla}) \cdot \mathbf{J}_{nv} - \sum_1^{Ni} (\boldsymbol{\nabla} \mu_{ni} + \Omega \boldsymbol{\Sigma} \cdot \boldsymbol{\nabla}) \cdot \mathbf{J}_{ni} \right] dV \geq 0. \end{aligned} \quad (16)$$

We can now decompose the above dissipation rate into two independent contributions as $d = d_D + d_P$, where d_D is due to diffusion, and d_P is associated with dislocation motion. If we further assume that near equilibrium, the dissipation d_D is quadratic in the flux, then the principle of maximum dissipation yields a linear relation between $\mathbf{J}_\star$ and the corresponding thermodynamic driving force $\boldsymbol{\nabla} \mu_\star - \Omega \boldsymbol{\Sigma} \cdot \boldsymbol{\nabla}$ (Onsager, 1931; Ziegler, 1958, 1963), with a tensor coefficient scaling as $c_\star/k_B T \exp(-E_\star^m/k_B T)$ (Larchet and Cahn, 1982), where $E_\star^m$ is the migration energy of a generic species $\star = ni, nv$. Localization then yields the conditions

$$\mathbf{J}_{nv} = -\frac{nc_{nv}}{\Omega k_B T} \mathbf{D}_{nv} (\boldsymbol{\nabla} \mu_{nv} - \Omega \boldsymbol{\Sigma} \cdot \boldsymbol{\nabla}) \quad \text{in } \mathcal{V}, \quad (17)$$

$$\mu_{nv} = g_{nv} + \Omega \hat{\mathbf{n}} \cdot \boldsymbol{\Sigma} \hat{\mathbf{n}} \quad \text{on } \partial\mathcal{V} \quad (18)$$

$$\mathbf{J}_{ni} = -\frac{nc_{ni}}{\Omega k_B T} \mathbf{D}_{ni} (\boldsymbol{\nabla} \mu_{ni} + \Omega \boldsymbol{\Sigma} \cdot \boldsymbol{\nabla}) \quad \text{in } \mathcal{V}, \quad (19)$$

$$\mu_{ni} = g_{ni} - \Omega \hat{\mathbf{n}} \cdot \boldsymbol{\Sigma} \hat{\mathbf{n}} \quad \text{on } \partial\mathcal{V}. \quad (20)$$

where $\mathbf{D}_\star = \mathbf{D}_{0\star} \exp(-E_\star^m/k_B T)$ is the temperature-dependent diffusion tensor of species $\star$, and $\mathbf{D}_{0\star}$ is a positive-definite pre-exponential tensor. We note that in the presence of elastic strain fields, the diffusion tensor need not remain constant, and anisotropy of the saddle-point configuration leads to strain-dependent corrections to diffusion, that are captured through an elastodiffusion tensor (Dederichs and Schroeder, 1978; Woo, 1984; Da Fonseca et al., 2023). It is also worth noting that a pre-factor n appears in the flux because the chemical potential is defined per point defect belonging to a n -size cluster.

Given these relations, the condition that the plastic velocity gradient $\mathbf{L}^P$ must satisfy is then

$$\int_{\mathcal{V}} d_P \, dV = \int_{\mathcal{V}} \left[\boldsymbol{\Sigma} : \mathbf{L}^P - \sum_1^{Nv} \mu_{nv} p_{nv} - \sum_1^{Ni} \mu_{ni} p_{ni} \right] dV \geq 0. \quad (21)$$

It can be seen that Eq. (21) is valid in general, regardless of the specific representation of the plastic flow. In particular for the case of a network $\mathcal{N}$ of discrete dislocations, the plastic velocity gradient $\mathbf{L}^P$ and the production terms $p_\star$ are concentrated on the dislocation loops, that is

$$\mathbf{L}^P(\mathbf{x}) = - \int_{\mathcal{N}} \delta(\mathbf{x} - \mathbf{x}') \mathbf{b}' \otimes \mathbf{w}' \times d\boldsymbol{\ell}' \text{ and } p_\star = \frac{1}{\Omega} \int_{\mathcal{N}} \delta(\mathbf{x} - \mathbf{x}') \mathbf{w}'_\star \cdot (\mathbf{b} \times d\boldsymbol{\ell}'), \quad (22)$$

where δ is the Dirac delta function, $\mathbf{b}$ being the local dislocation Burgers vector and $\mathbf{w}$ is the dislocation velocity vector. It should be noted that $\mathbf{w}_\star$ is the velocity at which the $\star$ cluster family is generated and $d\boldsymbol{\ell}'$ is an infinitesimal line element of the dislocation. We assume that the plastic dissipation is localized on the dislocation line and arises from two independent contributions, namely dissipation due to phonon scattering during dislocation glide, and dissipation related to defect emission/absorption at the dislocation. Then the climb component of the dislocation velocity can be written as $\mathbf{w}_c = \left(\sum_1^{Nv} w_{nv} - \sum_1^{Ni} w_{ni} \right) \boldsymbol{\gamma}$, where $\boldsymbol{\gamma} = \mathbf{b} \times \boldsymbol{\xi} / \|\mathbf{b} \times \boldsymbol{\xi}\|$ is the climb direction and $\boldsymbol{\xi}$ is the dislocation line direction, while the glide component of the velocity is $\mathbf{w}_g = \mathbf{w} - \mathbf{w}_c$. It is worth noting that the climb motion of dislocations is driven by the net flux of defects. Using the decomposition, a quadratic dissipation in the corresponding irreversible rates can therefore be written as:

$$d_P(\mathbf{x}) = \int_{\mathcal{N}} \delta(\mathbf{x} - \mathbf{x}') \left[B_g w_g'^2 + \sum_1^{Nv} A_{nv} w_{nv}'^2 + \sum_1^{Ni} A_{ni} w_{ni}'^2 \right] d\boldsymbol{\ell}'. \quad (23)$$

where B_g is the effective glide drag coefficient and the factor $A_\star$ characterizes the climb dissipation rate arising from the flux of the species $\star$. In combination with the Eq. (21), we have

$$\begin{aligned} \int_{\mathcal{N}} \left[B_g w_g'^2 + \sum_1^{Nv} A_{nv} w_{nv}'^2 + \sum_1^{Ni} A_{ni} w_{ni}'^2 \right] d\boldsymbol{\ell}' &= \int_{\mathcal{N}} \left[(\boldsymbol{\Sigma} \mathbf{b} \times \boldsymbol{\xi}) \cdot \mathbf{w} - \sum_1^{Nv} \frac{\mu_{nv}}{\Omega} (\mathbf{b} \times \boldsymbol{\xi}) \cdot \mathbf{w}_{nv} - \sum_1^{Ni} \frac{\mu_{ni}}{\Omega} (\mathbf{b} \times \boldsymbol{\xi}) \cdot \mathbf{w}_{ni} \right] d\boldsymbol{\ell}' \\ &= \int_{\mathcal{N}} \left\{ f_{PK} \cdot \left[\mathbf{w}_g + \left(\sum_1^{Nv} w_{nv} - \sum_1^{Ni} w_{ni} \right) \frac{\mathbf{b} \times \boldsymbol{\xi}}{\|\mathbf{b} \times \boldsymbol{\xi}\|} \right] - \sum_1^{Nv} \frac{\mu_{nv}}{\Omega} (\mathbf{b} \times \boldsymbol{\xi}) \cdot \mathbf{w}_{nv} - \sum_1^{Ni} \frac{\mu_{ni}}{\Omega} (\mathbf{b} \times \boldsymbol{\xi}) \cdot \mathbf{w}_{ni} \right\} d\boldsymbol{\ell}', \end{aligned} \quad (24)$$

where we have introduced the Peach–Koehler force $\mathbf{f}_{\text{PK}} = \boldsymbol{\Sigma} \mathbf{b} \times \boldsymbol{\xi}$ (Peach and Koehler, 1950) in the above equation. The principle of maximum dissipation can therefore be applied to Eq. (24) with respect to the independent variations of the glide and climb component of dislocation velocity (Po and Ghoniem, 2014). This finally yields the constitutive equations²

$$B_g \mathbf{w}_g = \mathbf{f}_{\text{PK}g} \quad \text{on } \mathcal{N}, \quad (26)$$

$$A_{nv} w_{nv} = \frac{\mathbf{f}_{\text{PK}} \cdot (\mathbf{b} \times \boldsymbol{\xi})}{\|\mathbf{b} \times \boldsymbol{\xi}\|} - \frac{\mu_{nv}}{\Omega} \|\mathbf{b} \times \boldsymbol{\xi}\| \quad \text{on } \mathcal{N}, \quad (27)$$

$$A_{ni} w_{ni} = -\frac{\mathbf{f}_{\text{PK}} \cdot (\mathbf{b} \times \boldsymbol{\xi})}{\|\mathbf{b} \times \boldsymbol{\xi}\|} - \frac{\mu_{ni}}{\Omega} \|\mathbf{b} \times \boldsymbol{\xi}\| \quad \text{on } \mathcal{N}. \quad (28)$$

In thermodynamic equilibrium, the dislocation velocity is zero. In conjunction with Eqs. (27)–(28), the equilibrium chemical potential of defect clusters on a non-screw dislocation can thus be expressed as:

$$\mu_{nv}^{\text{eq}} = -\mu_{ni}^{\text{eq}} = \frac{\Omega \mathbf{f}_{\text{PK}} \cdot (\mathbf{b} \times \boldsymbol{\xi})}{\|\mathbf{b} \times \boldsymbol{\xi}\|^2}. \quad (29)$$

We have now determined the relations between the internal irreversible rates and the plastic deformation process in terms of the defect flux and the dislocation motion. This yields a closed system of equations that we shall implement within the DDD framework in the next section.

3. Discrete dislocation dynamics implementation

3.1. The mechanics and diffusion BVPs

In this section, we present the implantation of the defect evolution framework developed in Section 2.2 within the DDD framework. We start by prescribing an explicit form of the free energy density as functions of the strain tensor $\mathbf{E}^E$ and the defect cluster concentrations c_{nv}/n and c_{ni}/n :

$$W(\mathbf{E}^E, c_{nv}/n, c_{ni}/n) = \frac{1}{2} \mathbb{C} \mathbf{E}^E : \mathbf{E}^E + \frac{1}{\Omega_0} \sum_1^{Nv} \frac{E_{nv}^f c_{nv}/n}{1 + c_{nv}/n} + \frac{k_B T}{\Omega_0} \sum_1^{Nv} \left[\frac{c_{nv}/n}{1 + c_{nv}/n} \ln \left(\frac{c_{nv}/n}{1 + c_{nv}/n} \right) + \frac{1}{1 + c_{nv}/n} \ln \left(\frac{1}{1 + c_{nv}/n} \right) \right] \\ + \frac{1}{\Omega_0} \sum_1^{Ni} \frac{E_{ni}^f c_{ni}/n}{1 + c_{ni}/n} + \frac{k_B T}{\Omega_0} \sum_1^{Ni} \left[\frac{c_{ni}/n}{1 + c_{ni}/n} \ln \left(\frac{c_{ni}/n}{1 + c_{ni}/n} \right) + \frac{1}{1 + c_{ni}/n} \ln \left(\frac{1}{1 + c_{ni}/n} \right) \right], \quad (30)$$

where $\mathbb{C}$ is the rank-four tensor of elastic moduli and $E_{\star}^f$ are formation energies for n -vacancies or interstitials clusters. Within the DDD framework, the kinematics is generally assumed to be linearized. A generic deformation gradient $\mathbf{F}^{\star}$ is expressed as the sum of the identity tensor $\mathbf{I}$ and the corresponding distortion tensor $\boldsymbol{\beta}^{\star}$. Each $\boldsymbol{\beta}^{\star}$ is assumed to be small. By neglecting body forces and considering isothermal conditions, as well as the steady-state approximation for defect cluster concentrations ($\dot{c}_{\star} = 0$), the stress in the body becomes

$$\boldsymbol{\sigma} = \mathbb{C} \boldsymbol{\varepsilon}^E = \mathbb{C} (\mathbf{u} \boldsymbol{\nabla} - \boldsymbol{\beta}^P - \boldsymbol{\beta}^D - \boldsymbol{\beta}^V). \quad (31)$$

Here $\boldsymbol{\beta}^V$ is the distortion caused by the defect relaxation volume. From Eq. (10), we see that $\dot{\boldsymbol{\beta}}^V = \dot{\mathbf{F}}^V \approx \mathbf{L}^V$, hence

$$\boldsymbol{\beta}^V = \sum_1^{Ni} \frac{\chi_{ni}^r}{n \Omega_0} c_{ni} + \sum_1^{Nv} \frac{\chi_{nv}^r}{n \Omega_0} c_{nv}. \quad (32)$$

It is worth noting that this distortion is typically several orders of magnitude smaller than the other distortions and can be neglected. Meanwhile, the distortion $\boldsymbol{\beta}^D$ is associated with the defect flux. From (11) we find $\dot{\boldsymbol{\beta}}^D = \dot{\mathbf{F}}^D \approx \mathbf{L}^D \approx -\Omega_0 (\sum_1^{Nv} \boldsymbol{\nabla} \mathbf{J}_{nv} - \sum_1^{Ni} \boldsymbol{\nabla} \mathbf{J}_{ni})$. In fact, we can define a diffusional displacement $\boldsymbol{\zeta}$ whose gradient is $\boldsymbol{\beta}^D$, i.e., $\boldsymbol{\beta}^D = \boldsymbol{\zeta} \boldsymbol{\nabla}$. Then this displacement can be written as:

$$\boldsymbol{\zeta} := -\Omega_0 \int_0^t \left(\sum_1^{Nv} \mathbf{J}_{nv} - \sum_1^{Ni} \mathbf{J}_{ni} \right). \quad (33)$$

On the other hand, $\boldsymbol{\beta}^P$ characterizes the plastic distortion, and its average rate can be write as

$$\langle \dot{\boldsymbol{\beta}}^P \rangle = - \int_{\mathcal{N}} \frac{\mathbf{b} \otimes \mathbf{w} \times d\boldsymbol{\ell}}{V}. \quad (34)$$

² With the help of w_c , Eq. (27) can also be written in terms of the climb velocity. Let us define a parameter $A_e = \frac{1}{\sum_1^{Nv} 1/A_{nv} + \sum_1^{Ni} 1/A_{ni}}$, in this case we have

$$B_c w_c = \mathbf{f}_{\text{PK}} \cdot \frac{\mathbf{b} \times \boldsymbol{\xi}}{\|\mathbf{b} \times \boldsymbol{\xi}\|} - \sum_1^{Nv} \mu_{nv} \frac{A_e}{A_{nv}} \frac{\|\mathbf{b} \times \boldsymbol{\xi}\|}{\Omega} + \sum_1^{Ni} \mu_{ni} \frac{A_e}{A_{ni}} \frac{\|\mathbf{b} \times \boldsymbol{\xi}\|}{\Omega} \quad (25)$$

where we have introduced the effective climb drag coefficient $B_c = A_e$. The force terms on the right-hand side are the Peach–Koehler force projected in the climb direction and the Bardeen–Herring chemical force (Bardeen and Herring, 1952), respectively.

With these relations, the mechanics problem aims to find the displacement field $\mathbf{u}$ such that, given the traction vector $\mathbf{t}$ on the Neumann boundary $\partial_N \mathcal{V}$ and the applied displacement $\bar{\mathbf{u}}$ on the Dirichlet boundary $\partial_D \mathcal{V}$, the following boundary value problem (BVP) is satisfied:

$$\begin{cases} \boldsymbol{\sigma} \cdot \boldsymbol{\nabla} = \mathbf{0} & \text{in } \mathcal{V} \\ \boldsymbol{\sigma} = \mathbb{C}[(\mathbf{u} - \boldsymbol{\zeta}) \boldsymbol{\nabla} - \boldsymbol{\beta}^P] & \text{in } \mathcal{V} \\ \boldsymbol{\sigma} \hat{\mathbf{n}} = \mathbf{t} & \text{on } \partial_N \mathcal{V} \\ \mathbf{u} = \bar{\mathbf{u}} & \text{on } \partial_D \mathcal{V} \end{cases} \quad (35)$$

On the other hand, this mechanics problem is coupled to the BVP governing the diffusion of mobile defects and defect clusters, which can be obtained from the balance of defects shown in Eq. (5). Recalling the expression of the production term Eq. (9), the boundary conditions Eqs. (18), (20), and the internal condition on dislocations Eq. (29), we have

$$\begin{cases} -\mathbf{J}_{nv} \cdot \boldsymbol{\nabla} + p_{nv} + F_{nv} = 0 & \text{in } \mathcal{V} \\ -\mathbf{J}_{ni} \cdot \boldsymbol{\nabla} + p_{ni} + F_{ni} = 0 & \text{in } \mathcal{V} \\ c_{nv} = c_{0nv} \exp\left(\frac{n g_{nv} + n f_n \Omega_0}{k_B T}\right) & \text{on } \partial \mathcal{V} \\ c_{ni} = c_{0ni} \exp\left(\frac{n g_{ni} - n f_n \Omega_0}{k_B T}\right) & \text{on } \partial \mathcal{V} \end{cases} \quad (36)$$

where we have introduced the reference vacancy concentration in the stressed crystal as

$$c_{0nv} := n \exp\left(-\frac{E_{nv}^f - \chi_{nv}^r : \boldsymbol{\sigma}}{k_B T}\right) \quad c_{0ni} := n \exp\left(-\frac{E_{ni}^f - \chi_{ni}^r : \boldsymbol{\sigma}}{k_B T}\right). \quad (37)$$

In addition, from the free energy shown in Eq. (30) and the definition of the chemical potential, the chemical potential per vacancy or interstitial in a n -size cluster is written as

$$\mu_{nv} = \frac{1}{n} \left[E_{nv}^f + k_B T \ln\left(\frac{c_{nv}}{n}\right) - \chi_{nv}^r : \boldsymbol{\sigma} \right] = \frac{1}{n} \left[E_{nv}^f + k_B T \ln\left(\frac{c_{nv}}{n}\right) - \mathbf{P}_{nv} : \boldsymbol{\epsilon}_{nv} \right] \quad (38)$$

$$\mu_{ni} = \frac{1}{n} \left[E_{ni}^f + k_B T \ln\left(\frac{c_{ni}}{n}\right) - \chi_{ni}^r : \boldsymbol{\sigma} \right] = \frac{1}{n} \left[E_{ni}^f + k_B T \ln\left(\frac{c_{ni}}{n}\right) - \mathbf{P}_{ni} : \boldsymbol{\epsilon}_{ni} \right]. \quad (39)$$

where we have introduced the elastic dipole tensors $\mathbf{P}_{nv}$ and $\mathbf{P}_{ni}$ of clusters containing n vacancies and interstitials, respectively, where the corresponding relaxation volumes are related to the dipole tensors by $\chi_{kl} = S_{kl ij} \mathbf{P}_{ij}$ (Dudarev and Ma, 2018), with $S_{kl ij}$ being the elastic compliance tensor. It is noteworthy that the isotropic and deviatoric components of the dipole tensor are respectively associated to the size and shape effects of defects. It is also interesting to note that as $E_\star^f$ is the formation energy of a n -size defect cluster, $E_\star^f/n$ thus represents the formation energy per defect in such a cluster. This is self-consistent with the definition of the defect concentration in the n -size clusters $c_\star$ and the defect cluster concentration $c_\star/n$. From this expression and the fact that stress is divergenceless, the defect flux in Eqs. (17) and (19) become

$$\mathbf{J}_{nv} = -\mathbf{D}_{nv} \left[\frac{1}{\Omega_0} \boldsymbol{\nabla} c_{nv} - \frac{c_{nv}}{\Omega_0 k_B T} \boldsymbol{\nabla} (\mathbf{P}_{nv} : \boldsymbol{\epsilon}_{nv}) \right] \quad (40)$$

$$\mathbf{J}_{ni} = -\mathbf{D}_{ni} \left[\frac{1}{\Omega_0} \boldsymbol{\nabla} c_{ni} - \frac{c_{ni}}{\Omega_0 k_B T} \boldsymbol{\nabla} (\mathbf{P}_{ni} : \boldsymbol{\epsilon}_{ni}) \right]. \quad (41)$$

Substituting above equations into BVPs, Eqs. (35) and (36) can be solved with the help of the superposition principle.

3.2. Superposition applied to BVPs

The BVP (35) is solved by a generalization of the superposition principle proposed by Van der Giessen and Needleman (1995). The total displacement field in the body is decomposed as

$$\mathbf{u} = \mathbf{u}^\infty + \tilde{\mathbf{u}} + \boldsymbol{\zeta}, \quad (42)$$

where $\mathbf{u}^\infty$ is the infinite-medium displacement caused by dislocations, $\boldsymbol{\zeta}$ is the diffusive displacement introduced in (33) and $\tilde{\mathbf{u}}$ is the finite-medium displacement correction. The infinite-medium displacement $\mathbf{u}^\infty$ and the finite-medium correction $\tilde{\mathbf{u}}$ solve the following sub-problems, respectively,

$$\begin{cases} \boldsymbol{\sigma}^\infty \cdot \boldsymbol{\nabla} = \mathbf{0} & \text{in } \mathbb{R}^3 \\ \boldsymbol{\sigma}^\infty = \mathbb{C}(\mathbf{u}^\infty \boldsymbol{\nabla} - \boldsymbol{\beta}^P) & \text{in } \mathbb{R}^3 \end{cases}, \quad \text{and} \quad \begin{cases} \boldsymbol{\sigma}^\star \cdot \boldsymbol{\nabla} = \mathbf{0} & \text{in } \mathcal{V} \\ \tilde{\boldsymbol{\sigma}} = \mathbb{C} \tilde{\mathbf{u}} \boldsymbol{\nabla} & \text{in } \mathcal{V} \\ \boldsymbol{\sigma}^\star \hat{\mathbf{n}} = \mathbf{t} - \boldsymbol{\sigma}^\infty \hat{\mathbf{n}} & \text{on } \partial_N \mathcal{V} \\ \tilde{\mathbf{u}} = \bar{\mathbf{u}} - \mathbf{u}^\infty - \boldsymbol{\zeta} & \text{on } \partial_D \mathcal{V} \end{cases}. \quad (43)$$

The total stress field in the body is $\boldsymbol{\sigma} = \boldsymbol{\sigma}^\infty + \tilde{\boldsymbol{\sigma}}$. This decomposition of the solution allows us to find the dislocation fields $\mathbf{u}^\infty$ and $\tilde{\boldsymbol{\sigma}}$ by the standard method of Green's functions, typically employed within the DDD framework. The finite-medium correction BVP is solved using the Galerkin Finite Element Method (FEM), concurrently coupled to the DDD simulation.

In order to solve the BVP (36), the defect concentration of the species $\star$ is also additively decomposed as

$$c_\star = c_\star^\infty + \tilde{c}_\star, \quad (44)$$

where $c_\star^\infty$ is the defect concentration field generated by dislocations climbing in the infinite medium, while $\tilde{c}_\star$ is the finite-medium concentration correction. We can concurrently additively decompose the flux of the species $\star$, from Eq. (40) and (41) as

$$\mathbf{J}_\star = -D_\star \underbrace{\left[\frac{1}{\Omega_0} \nabla c_\star^\infty + \frac{c_\star}{\Omega_0 k_B T} \nabla (\mathbf{P}_\star : \boldsymbol{\varepsilon}_\star) \right]}_{\mathbf{J}_\star^\infty} - D_\star \underbrace{\left[\frac{1}{\Omega_0} \nabla \tilde{c}_\star \right]}_{\tilde{\mathbf{J}}_\star}, \quad (45)$$

The second term in the flux expression, $\mathbf{J}_\star^\infty$, characterizes the drift of mobile point defects due to their elastic interaction with sinks (dislocations), which contributes to the so-called dislocation bias (Wolfer, 2007). Importantly, however, two distinct elastic effects contribute to the bias: (1) the aforementioned drift term, which arises from the gradient of the interaction energy evaluated in the equilibrium configuration, directly altering the defect flux; and (2) dependence of the diffusion tensor on strain, arising from the anisotropic modification of the saddle-point energy, thus modifying the migration barrier and effective diffusivity (Dederichs and Schroeder, 1978). In the Green function based formulation used here, both effects are absorbed into a scalar bias factor, while the anisotropy of the diffusion tensor is considered explicitly. Note that the concentration of $c_\star^\infty$ is associated with the flux toward the dislocation; therefore, the total flux is approximated by

$$\mathbf{J}_\star \approx - \underbrace{\frac{D_\star}{\Omega_0} Z_\star \nabla c_\star^\infty}_{\mathbf{J}_\star^\infty} - \underbrace{\frac{D_\star}{\Omega_0} \nabla \tilde{c}_\star}_{\tilde{\mathbf{J}}_\star}, \quad (46)$$

where $Z_\star$ is the dislocation bias. The infinite-medium concentration $c_\star^\infty$ and the finite-medium concentration correction $\tilde{c}_\star$ solve the following sub-problems, respectively,

$$\begin{cases} -\mathbf{J}_\star^\infty \cdot \nabla + p_\star = 0 & \text{in } \mathbb{R}^3 \\ \mathbf{J}_\star^\infty = -Z_\star / \Omega_0 D_\star \nabla c_\star^\infty & \text{in } \mathbb{R}^3 \end{cases}, \quad \text{and} \quad \begin{cases} -\tilde{\mathbf{J}}_\star \cdot \nabla + F_\star = 0 & \text{in } \mathcal{V} \\ \tilde{\mathbf{J}}_\star = -1/\Omega_0 D_\star \nabla \tilde{c}_\star & \text{in } \mathcal{V} \\ \tilde{c}_\star = c_{0\star} \exp\left(\frac{n g_\star \pm n n_\star \Omega_0}{k_B T}\right) - c_\star^\infty & \text{on } \partial \mathcal{V} \end{cases}. \quad (47)$$

Similar to the mechanics problem, this decomposition is designed so that the infinite-medium concentration is determined by the Green's function method, while the finite-medium concentration correction is obtained through a Galerkin FEM solution.

Eliminating $\mathbf{J}_\star^\infty$ from the infinite-medium problem shows that the concentration $c_\star^\infty$ solves the anisotropic diffusion equation $(D_\star \nabla c_\star^\infty) \cdot \nabla + \Omega_0 p_\star / Z_\star = 0$, and therefore admits the solution $c_\star^\infty = -G \ast \Omega_0 p_\star / Z_\star$, where G is the Green's function of the anisotropic Poisson's equation with tensor coefficient $D_\star$, which can be obtained from its isotropic counterpart by a simple change of variables. Taking the expression of $p_\star$ from Eq. (22), we thus have

$$c_\star^\infty(\mathbf{x}) = \int_{\mathcal{N}} \frac{\mathbf{w}'_\star \cdot (\mathbf{b}' \times d\boldsymbol{\ell}')}{4\pi \sqrt{\det(Z_\star D_\star)} (\mathbf{x} - \mathbf{x}') \cdot (Z_\star D_\star)^{-1} (\mathbf{x} - \mathbf{x}')} d\boldsymbol{\ell}'. \quad (48)$$

3.3. Dislocation climb

The analytical expression of $c_\star^\infty$ can be found in Appendix A. When the climb velocity is small, defects are assumed to remain in equilibrium with dislocations. Rewriting (29) in terms of concentration, this means that the vacancy concentration along the dislocations is set to the equilibrium value

$$c_{nv}^{\text{eq}} = c_{0nv} \exp \left[\frac{n \Omega f_{PK} \cdot (\mathbf{b} \times \boldsymbol{\xi})}{k_B T \|\mathbf{b} \times \boldsymbol{\xi}\|^2} \right] c_{ni}^{\text{eq}} = c_{0ni} \exp \left[-\frac{n \Omega f_{PK} \cdot (\mathbf{b} \times \boldsymbol{\xi})}{k_B T \|\mathbf{b} \times \boldsymbol{\xi}\|^2} \right] \text{ on dislocations}. \quad (49)$$

Within the superposition scheme, the condition (49) for each species reads $c_\star^\infty(\mathbf{x}) + \tilde{c}_\star(\mathbf{x}) = c_\star^{\text{eq}}(\mathbf{x})$. Using (48), this condition can be rewritten as

$$\int_{\mathcal{N}} \frac{\mathbf{w}'_\star \cdot (\mathbf{b}' \times d\boldsymbol{\ell}')}{4\pi \sqrt{\det(Z_\star D_\star)} (\mathbf{x} - \mathbf{x}') \cdot (Z_\star D_\star)^{-1} (\mathbf{x} - \mathbf{x}')} = c_\star^{\text{eq}}(\mathbf{x}) - \tilde{c}_\star(\mathbf{x}). \quad (50)$$

Finally, the dislocation climb velocity is simply

$$\mathbf{w}_c = \sum_1^{Nv} \mathbf{w}_{nv} - \sum_1^{Ni} \mathbf{w}_{ni}. \quad (51)$$

Note that the above expressions account for contributions from all types of mobile defect clusters, dislocation bias effects, and diffusional anisotropy. For the isotropic diffusion case, the condition reduces to

$$\frac{1}{4\pi Z_\star D_\star} \int_{\mathcal{N}} \frac{\mathbf{w}'_\star \cdot (\mathbf{b}' \times d\boldsymbol{\ell}')}{\|\mathbf{x} - \mathbf{x}'\|} = c_\star^{\text{eq}}(\mathbf{x}) - \tilde{c}_\star(\mathbf{x}). \quad (52)$$

Table 1

Parameters used in the model validation simulations for the irradiation damage production, and mobile species properties.

Definition	Symbol	Value	Reference/Comment
Damage Production			
Temperature	T	1000 K	Carpenter et al. (1981, 1988)
Dose rate	G_0	0, 10^{-5} dpa/s	Carpenter et al. (1981, 1988)
Fraction of recombining point defects	f_r	0.99	Zhou et al. (2018), Kim et al. (2018)
Mobile Species			
Relative relaxation volume of vacancy	$\text{tr}(\chi_v^r)/\Omega_0$	-0.405	Varvenne et al. (2014), Le Bacq et al. (1999)
Relative relaxation volume of single interstitial	$\text{tr}(\chi_i^r)/\Omega_0$	1.2	Vérité et al. (2013)
Formation energy of vacancy	E_v^f	1.5 eV	Ouyang et al. (2018), Hood (1986)
Formation energy of interstitial	E_i^f	3.0 eV	Ouyang et al. (2018), Peng et al. (2012)
Migration energies of vacancy	$E_v^{m(11,22,33)}$	0.9 eV	Osetsyky et al. (2002), Dai et al. (2009)
Migration energies of interstitial	$E_i^{m(11,22,33)}$	0.09 eV	Mikhin et al. (1994), Osetsyky et al. (2002)
Diffusion coefficient prefactor of vacancy	D_{v0}	4.9639×10^{-10} m ² /s	Li et al. (2020), Christien and Barbu (2009)
Diffusion coefficient prefactor of single interstitial	D_{i0}	8.1882×10^{-10} m ² /s	Li et al. (2020), Christien and Barbu (2009)
Interaction bias factor of vacancies	$Z_{(c,a)}^v$	1.0	Woo (1988), Wolfer (2007)
Interaction bias factor of single interstitials	$Z_{(c,a)}^i$	1.0	Woo (1988), Wolfer (2007)

Accordingly, the dislocation climb velocity is determined by the net flux of defects, which reads

$$\int_{\mathcal{N}} \frac{\mathbf{w}_c' \cdot (\mathbf{b}' \times d\boldsymbol{\ell}')}{4\pi \|\mathbf{x} - \mathbf{x}'\|} = \sum_i^{Nv} Z_{nv} [D_{nv} c_{nv}^{\text{eq}}(\mathbf{x}) - D_{nv} \tilde{c}_{nv}(\mathbf{x})] - \sum_i^{Ni} Z_{ni} [D_{ni} c_{ni}^{\text{eq}}(\mathbf{x}) - D_{ni} \tilde{c}_{ni}(\mathbf{x})]. \quad (53)$$

4. Results

We now present applications of the proposed framework to dislocation climb in Zr, based on an implementation in the Mechanics of Defects Evolution Library (MoDELlib2, 2025). The effects of the external force and the coupled climb-glide motion of dislocations can be found in our previous work (Po et al., 2022). In the present work, no traction is applied to the boundary to simulate irradiation growth, and dislocations are allowed to climb only. The numerical implementation is in a Cartesian frame, and lattice vectors can be converted to the Miller–Bravais representation as: $\mathbf{a}_1 = a\mathbf{e}_x$, $\mathbf{a}_2 = -\frac{1}{2}a\mathbf{e}_x + \frac{\sqrt{3}}{2}a\mathbf{e}_y$, $\mathbf{a}_3 = -\frac{1}{2}a\mathbf{e}_x - \frac{\sqrt{3}}{2}a\mathbf{e}_y$, $\mathbf{c} = c\mathbf{e}_z$. A hexagonal unit cell with $c/a = \sqrt{8/3}$ is adopted, which slightly deviates from the actual ratio of Zr (1.59), and would result in a negligible anisotropy of about 2.7% along the c -axis (Monnet et al., 2004). We assume Zr as an elastically isotropic material with shear modulus $\mu = 33$ GPa, and Poisson's ratio $\nu = 0.34$ (Li et al., 2023).

4.1. Computational model validation

To validate the computational framework, we first consider the production of single interstitials and vacancies, as well as their interaction with a single isolated vacancy or interstitial dislocation loop. We account for the reaction between vacancy and interstitial defects following the spatially resolved cluster dynamics model developed in previous work (Maron et al., 2024), which yield the reaction terms in Eq. (4) as $F_i = -\alpha_{iv}c_i c_v + G_i$ and $F_v = -\alpha_{iv}c_i c_v + G_v$, with α_{iv} being the recombination rate. A single vacancy ($\langle c \rangle$) loop ($\text{tr}(\langle \beta_p \rangle) < 0$) with basal habit plane and $\mathbf{b} = c/2[0001]$, or a single interstitial ($\langle a \rangle$) loop ($\text{tr}(\langle \beta_p \rangle) > 0$) with 2nd order prismatic habit plane and $\mathbf{b} = a/3[2\bar{1}10]$ are placed at the center of the simulation volume with an initial radius of 40 nm. The simulation volume is a cubic box with a side length of $L = 3093|\mathbf{b}|$, where $|\mathbf{b}| = 0.3233$ nm is the magnitude of the Burgers vector for Zr. The simulation box size is much larger than the loop size, thereby the boundary presents no effect on the loop evolution as discussed in Appendix E. Only single interstitials and single vacancies are considered, with isotropic diffusion and no bias to the dislocations. The temperature is set to be 1000 K, and the dose rates chosen are 0 dpa/s for the thermal case, and 10^{-5} dpa/s for fast reactor conditions.

The properties of vacancies and interstitials are derived from experiments or empirical potentials for hcp zirconium, as summarized in Table 1. The fraction of defects that survive from displacement cascades depends on the primary knock on energy and temperature, but is in general below 5% (Zhou et al., 2018; Kim et al., 2018). Experimental evidence and atomistic simulations suggest that the formation energy of vacancies is approximately 1.5 eV (Hood, 1986; Mikhin et al., 1994; Wooding et al., 1998). Vacancy migration has generally been found to be close to isotropic, with a migration energy between 0.6–1.0 eV (Osetsyky et al., 2002; Mikhin et al., 1994; Kim et al., 2006; Dai et al., 2009). Here, we assume an isotropic vacancy diffusion with an energy barrier of 0.9 eV. The relative relaxation of vacancies is taken to be -0.405 (Varvenne et al., 2014; Le Bacq et al., 1999). The diffusion coefficient of vacancies is generally unclear from experiments and atomistic simulations, where the pre-exponential term has large deviations; here we adopt values close to experimental values, and a discussion on the vacancy diffusion coefficient can be found in Appendix B. On the other hand, the most stable site for interstitials, identified through empirical potentials and DFT, is typically the basal octahedral site, and exhibit formation energies around 3.0 eV (Peng et al., 2012; Ouyang et al., 2018; Vérité et al., 2013; Varvenne et al., 2013). Interstitials can diffuse anisotropically with faster migration in the basal plane than along the basal pole, although the values for migration energies vary depending upon the interstitial site considered, and the atomistic method

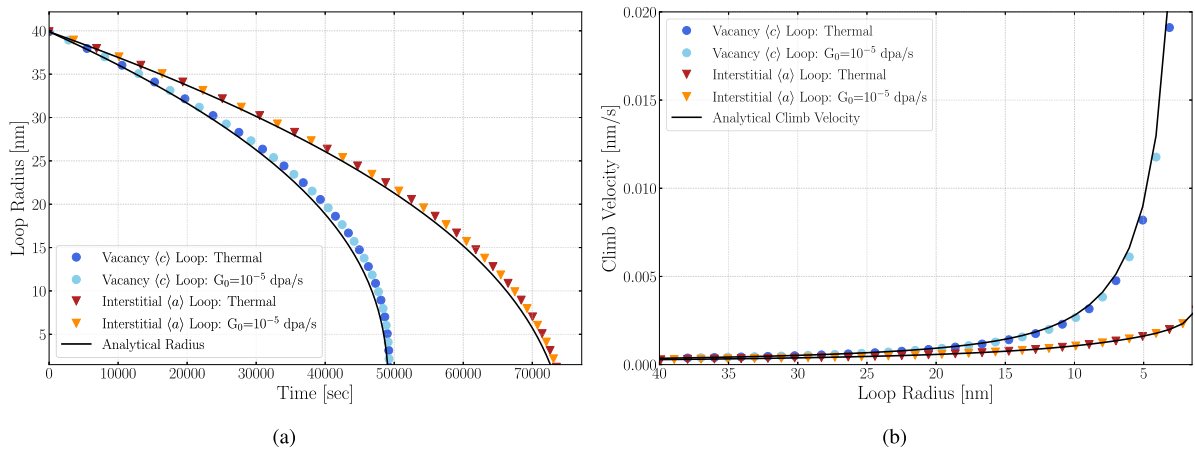


Fig. 2. Computational model validation. (a) Interstitial (a) loop (denoted by $\blacktriangledown$) and vacancy (c) loop (denoted by $\bullet$) radius evolution under isotropic and unbiased conditions for thermal ($G_0 = 0$ dpa/s), and fast reactor conditions ($G_0 = 10^{-5}$ dpa/s), and compared with the analytical solution. (b) Interstitial (a) loop and vacancy (c) loop average nodal climb velocity as a function of loop radius compared with the analytical solution of the climb velocity for thermal conditions, and fast reactor conditions.

used (Ghorbani et al., 2024). Typically, the migration energies are an order of magnitude smaller than that of the vacancies, and here we assume that interstitials diffuse isotropically with a migration energy of 0.09 eV. A relaxation volume of $1.2\Omega_0$ is used for interstitials (Vérité et al., 2013).

The results of the model validation study are shown in Fig. 2. We compare the model developed here with the analytical solution for the climb velocity of a prismatic dislocation loop in the presence of a generation of point defects as shown in Appendix C. From Fig. 2(a) we find that both the interstitial (a) and vacancy (c) dislocation loops shrink regardless of dose rate ($G_0 = 0, 10^{-5}$ dpa/s). Although individual contributions to the climb velocity from vacancies and interstitials depend on the production rate (Eq. (C.19) & Eq. (C.20)), the net climb velocity does not (Eq. (C.23)). Therefore, regardless of dose rate, in an unstressed crystal, with isotropic diffusion and no dislocation bias, the dislocation loops will act as the thermal case (Was, 2007). The climb velocity is then proportional to the difference between the thermal vacancy and the SIA flux to the core, $D_v c_{0v} - D_i c_{0i}$, which depends solely on the formation energy and migration energy of the interstitial and vacancy defects. Since $E_i^f \gg E_v^f$, the contribution of interstitials to the climb of dislocations can be neglected. The vacancy loop shrinks by emitting vacancies, as the equilibrium concentration for vacancies at the dislocation is higher than the thermal concentrations, which has been shown for the annealing of vacancy loops in aluminum by Po et al. (2022). The interstitial loop shrinks through the absorption of vacancies, as the equilibrium concentration for vacancies at the dislocation is smaller than the thermal vacancy concentrations. In Fig. 2(b), the average net climb velocity of the dislocation loop nodes from the numerical model is also compared with the analytical expression from Eq. (C.22). We find a good agreement between the calculated dislocation nodal velocities and the analytical solution of the climb velocity.³

To better understand the evolution mechanism for a single loop, we extend the study to investigate the effects of anisotropic diffusion of interstitials, and defect reactions, on the concentration profiles of mobile defects in the finite crystal. A single interstitial loop with $b = a/3[2\bar{1}10]$ and a radius of 10 nm is placed in the center of the simulation domain. The concentration profiles along $[01\bar{1}0]$ and $[0001]$ without the reactions ($\alpha_{iv} = 0$) between interstitials and vacancies are shown in Fig. 3, for isotropic diffusion (solid curves) and anisotropic diffusion (dashed curves) of interstitials. The field plots of Figs. 3(a) & 3(b) correspond solely to the anisotropic case. For the isotropic case, the concentration field is highly symmetric in all directions. The background concentration profiles for interstitials and vacancies (cyan curves in Figs. 3(c)–3(f)) are a parabola, whose shape depends on the dose rate, diffusivity, simulation size, and the boundary condition at the surfaces, i.e., $c_{0v} = e^{-E_v^f/kT}$ and $c_{0i} = e^{-E_i^f/kT}$. As the equilibrium vacancy and interstitial concentrations c_v^{eq} and c_i^{eq} on dislocations are much lower than the background concentrations due to the supersaturation of defects from the irradiation cascade, both defects move towards dislocations, leading to a negative dislocation concentration profile c^∞ . Note that the interstitial flux induces an opposite contribution to the dislocation climb velocity compared to its vacancy counterpart. The actual dislocation motion is thus a result of their net flux and is independent on the local supersaturation when no diffusional anisotropy or dislocation bias is considered, discussed in Appendix C. The climb velocity of Eq. (53) reduces to the thermal case, and the interstitial loop will shrink.

If interstitials diffuse anisotropically, still without the reactions ($\alpha_{iv} = 0$), the concentration profiles change for interstitials (Figs. 3(d) & 3(f)). Here we take $E_i^{m(33)} = 0.158$ eV along $[0001]$ direction, and $E_i^{m(11)} = E_i^{m(22)} = 0.056$ eV along $[2\bar{1}10]$ and $[01\bar{1}0]$

³ The non-singular theory from Cai et al. (2006) was used in the DDD model and for the Peach–Koehler force calculation in the analytical solution. For simplicity, the hydrostatic component of the stress tensor was excluded from the calculation of the thermal equilibrium concentration for direct comparison. In the model we retain the full stress tensor, including the hydrostatic term, to accurately capture the response of stress fields.

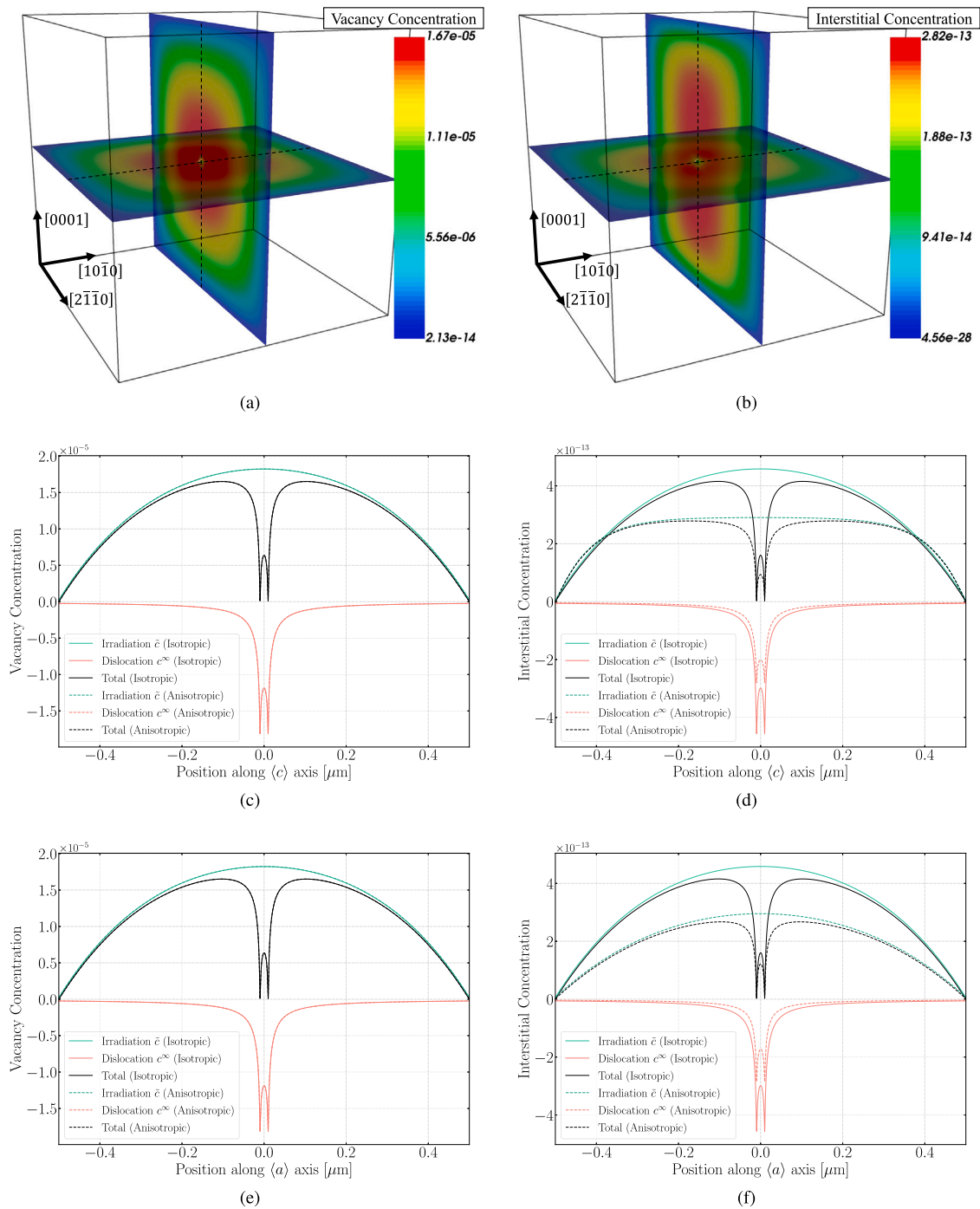


Fig. 3. Total concentration field plots for anisotropic interstitials & concentration line plots for isotropic (solid lines) and anisotropic (dashed lines) interstitials of the individual contributions of $\tilde{c}_*$ & c_*^∞ to c_*^{Total} for vacancy and interstitial defects along the $\langle a \rangle$ and $\langle c \rangle$ crystallographic directions **without reactions**. (a) Field plot of the total vacancy concentrations (c_v^{Total}) in the basal plane and along the basal pole for anisotropic interstitials ($p^m = 0.7$). (b) Field plot of the total interstitial concentrations (c_i^{Total}) in the basal plane and along the basal pole for anisotropic interstitials ($p^m = 0.7$). (c) Concentration profile of vacancies along $\langle c \rangle$ axis. (d) Concentration profile of interstitials along $\langle c \rangle$ axis. (e) Concentration profile of vacancies along $\langle a \rangle$ axis. (f) Concentration profile of interstitials along $\langle a \rangle$ axis.

directions. Near the boundaries, the total concentrations still approach c_{0v} and c_{0i} due to the presence of free surfaces. The DAD does not affect the concentration profile of the vacancies along [0001] or [0110]; however, the DAD has lowered the interstitial

Table 2Values of p^m and corresponding $E^{m(33)}$ and $E^{m(11,22)}$ for di-interstitials.

Anisotropy factor	p^m	1.0	0.9	0.8	0.7	0.6	0.5
Migration energies	$E_{2i}^{m(33)}$ (eV)	0.090	0.110	0.132	0.158	0.187	0.222
Migration energies	$E_{2i}^{m(11,22)}$ (eV)	0.090	0.080	0.069	0.056	0.041	0.024

concentration $\tilde{c}_i$ along [0001] and [01 $\bar{1}$ 0] (Figs. 3(d) & 3(f)). Due to a smaller diffusivity, we see a nearly constant interstitial concentration in the center region of the crystal ($-0.4 \mu\text{m}$ to $0.4 \mu\text{m}$) along [0001] direction, and a steep gradient near the boundaries with a thickness of ~ 100 nm. The equilibrium concentrations of the dislocation line are unchanged and still lower than $\tilde{c}$ and the dislocation concentration c^∞ remains negative; however, it is smaller than that of the isotropic case. Thus, the flux of vacancies to the loop is larger than interstitials, and from Eq. (53), the loop will shrink more rapidly than in the thermal case. It is also noteworthy that the interstitial concentrations (cyan dashed curves) are not the same along the [0001] and [01 $\bar{1}$ 0] directions, leading to climb velocity variation along differently oriented segments of the dislocation, naturally resulting in an elliptical shape. It is emphasized that the change of the loop shape will re-distribute the Peach–Koehler force, influencing the shrinking/growing rate of the loop (Woo, 1988; Saidi et al., 2020).

We now repeat this study with isotropic or anisotropic interstitials, *including the reactions* of interstitial and vacancy defects. The concentration profiles including recombination are shown in Fig. 4, again along [0001] and [01 $\bar{1}$ 0]. Comparing the isotropic cases of Fig. 3 with Fig. 4, we see that the main effect of recombination is to lower the maximum irradiation concentration $\tilde{c}_v$ and $\tilde{c}_i$ along both axes, from $\approx 1.8 \times 10^{-5}$ to $\approx 7.5 \times 10^{-7}$ for vacancies, and $\approx 4.1 \times 10^{-13}$ to $\approx 2.0 \times 10^{-14}$ for interstitials. In addition, the concentration profile in the center region of the crystal becomes nearly flat, leaving a 100 nm boundary layer with a steeper gradient to the boundary concentration. However, looking at Figs. 4(a) & 4(b), the inclusion of interstitial DAD significantly alters the concentration profiles. Along the $\langle a \rangle$ axis (Fig. 4(f)), the interstitial concentration slightly decreases due to the increase of its diffusivity in the basal plane. Along the $\langle c \rangle$ axis, however (Fig. 4(d)), the interstitial concentration is less than the isotropic case in the center of the crystal, but peaks near the top and bottoms of the basal pole. Due to slower diffusion along the $\langle c \rangle$ axis, the interstitial boundary layer is narrower with a steeper concentration gradient. The difference in the gradients of vacancies and interstitials near the boundary thus causes a spike of the interstitial concentration. When moving towards the center of the crystal, the concentration of vacancies begins to increase steadily, promoting more recombination. Because the vacancy and interstitial concentrations are now coupled with their recombination, the anisotropy acts to reduce the interstitial concentrations in the center region of the crystal, resulting in a higher vacancy concentration and a lower interstitial concentration compared with the isotropic case. Along the $\langle c \rangle$ axis (Figs. 4(a) & 4(b)), an anti-correlation of interstitial and vacancy concentrations is clearly evident.

4.2. The effect of bias

We next study the effects of different types of bias on the shrinkage and/or growth rates of the vacancy $\langle c \rangle$ and interstitial $\langle a \rangle$ dislocation loops. As discussed in the introduction, the reaction kinetics can be affected by various different biases in irradiated Zr, including dislocation bias, diffusion anisotropic difference (DAD), production bias (PB) and stress-induced preferential absorption (SIPA) bias (Woo, 2005). A brief discussion on various bias factors can be found in Appendix D. In the current model, we have incorporated the DAD and the dislocation bias by generalizing the Greene's function formalism to finite and anisotropic crystals, implemented using a Galerkin scheme suited for the nodal DDD method, as shown in Eq. (50). Production bias is included by considering the generation of point defects and di-interstitials from the cascade, such that the fractions of single vacancies, single interstitials and di-interstitials are characterized by f_v , f_i and f_{2i} respectively, and superposition is used to determine the contributions of each specie to the climb of the dislocation segments. The simulation setups are the same as those in the previous section, and follow Table 1, including:

1. The prismatic and basal loops consist of pure edge dislocation segments, and a dislocation bias of interstitial defects $Z_{\langle c,a \rangle}^{i,2i} > 1$ is applied to all segments, while the bias for vacancies is always kept at $Z_{\langle c,a \rangle}^v = 1$.
2. Cascades produce only single vacancies, single interstitials, and di-interstitials, since the binding energy of small vacancy clusters is low (Varvenne et al., 2014), and large interstitial clusters tend to immobilize.
3. Single vacancies and interstitials diffuse isotropically, with the migration energy $E_v^{m(11,22,33)} = 0.9$ eV, and $E_i^{m(11,22,33)} = 0.09$ eV, while di-interstitials diffuse anisotropically, characterized by the diffusion anisotropy factor p^m .⁴ The anisotropy factor is adjusted for the di-interstitials, keeping the effective diffusion coefficient $D_{eff}^i = [(D_{\langle a \rangle}^i)^2 \cdot D_{\langle c \rangle}^i]^{1/3}$ unchanged, as shown by Table 2.
4. Along with recombination of interstitial and vacancy defects, we include the interactions involving di-interstitials into Eq. (4), namely $F_v = -\alpha_{iv}c_ic_v - \beta_v^{2i}c_{2i}c_v + G_0$, $F_i = -\alpha_{iv}c_ic_v - \beta_i^i c_i c_i + \beta_v^{2i}c_{2i}c_v + G_0 f_i$ and $F_{2i} = -\beta_v^{2i}c_{2i}c_v + \beta_i^i c_i c_i + G_0 f_{2i}$, with β_v^{2i} and β_i^i being the corresponding reaction rates of defects (Maron et al., 2024).

⁴ The anisotropy factor is defined as $p^m = (\frac{D_{\langle c \rangle}^i}{D_{\langle a \rangle}^i})^{1/6}$, where $D_{\langle c \rangle}^i$ is the diffusion coefficient along the basal pole, and $D_{\langle a \rangle}^i$ is the diffusion coefficient in the basal plane (Woo, 1988; March-Rico and Wirth, 2022; Sakaël et al., 2025).

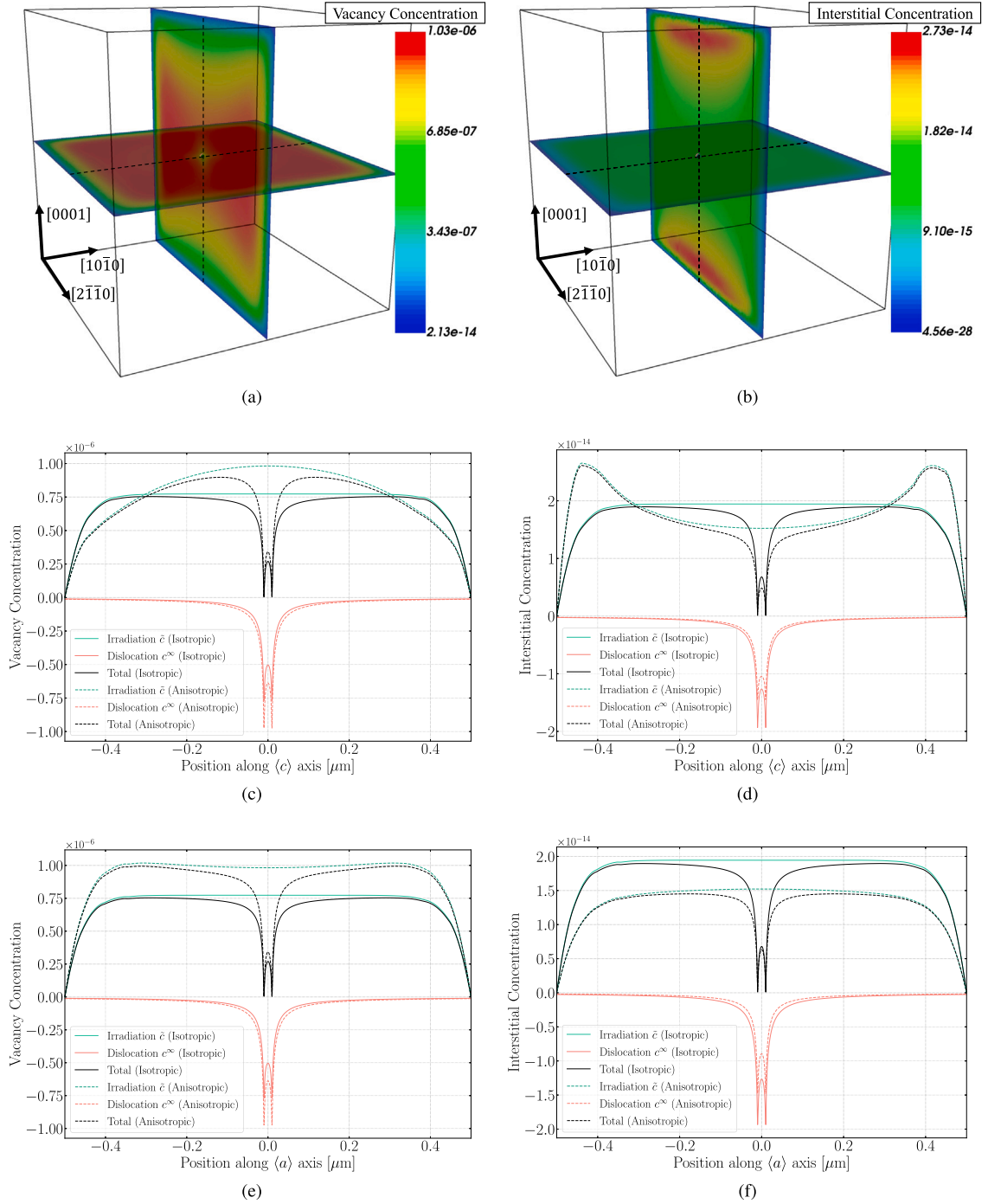


Fig. 4. Total concentration field plots for anisotropic interstitials & concentration line plots for isotropic (solid lines) and anisotropic (dashed lines) interstitials of the individual contributions of $\tilde{c}_*$ & c^∞ to c_v^{Total} for vacancy and interstitial defects along the $\langle a \rangle$ and $\langle c \rangle$ crystallographic directions **with reactions**. (a) Field plot of the total vacancy concentrations (c_v^{Total}) in the basal plane and along the basal pole for anisotropic interstitials ($p^m = 0.7$). (b) Field plot of the total interstitial concentrations (c_i^{Total}) in the basal plane and along the basal pole for anisotropic interstitials ($p^m = 0.7$). (c) Concentration profile of vacancies along $\langle c \rangle$ axis. (d) Concentration profile of interstitials along $\langle c \rangle$ axis. (e) Concentration profile of vacancies along $\langle a \rangle$ axis. (f) Concentration profile of interstitials along $\langle a \rangle$ axis.

We place a vacancy $\langle c \rangle$ or an interstitial $\langle a \rangle$ dislocation loop with a radius of $r = 20$ nm in the center of a simulation volume of side length $L = 3093|b|$, where $|b|$ is the magnitude of the Burgers vector in Zr, to eliminate boundary effects as discussed in

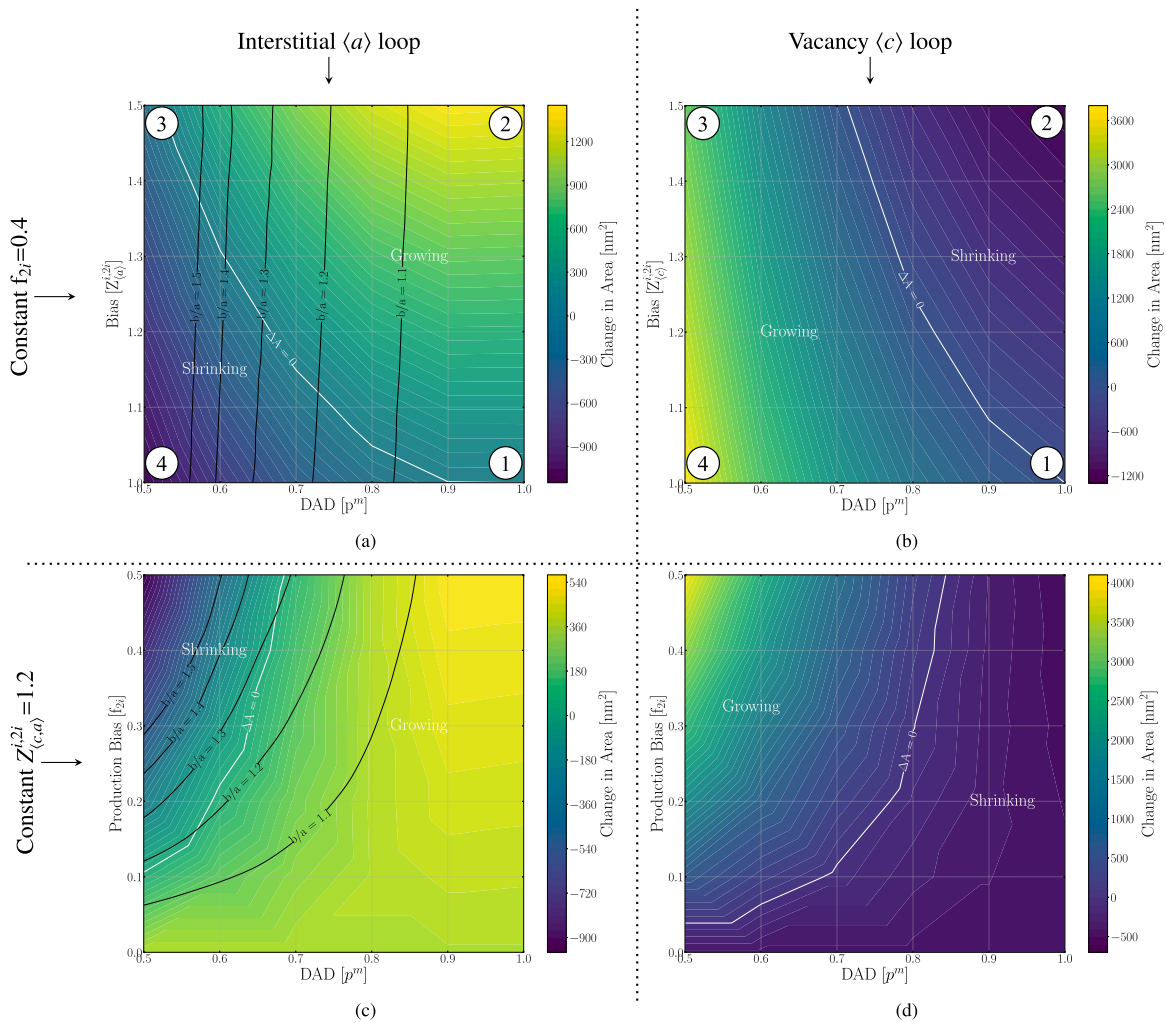


Fig. 5. The effects of dislocation bias, $Z_{(c,a)}^{i,2i}$, diffusion anisotropy difference, p^m , and production bias, f_{2i} , on the change of area of an individual vacancy & interstitial dislocation loop of $r = 20$ nm after 0.1 dpa. (a) The effects of DAD and dislocation bias on the growth rate of a prismatic interstitial loop with $f_{2i} = 0.4$. (b) The effects of DAD and dislocation bias on the growth rate of a basal vacancy loop with $f_{2i} = 0.4$. (c) The effects of DAD and production bias on the growth rate of a prismatic interstitial loop with $Z_{(c,a)}^{i,2i} = 1.2$. (d) The effects of DAD and production bias on the growth rate of a basal vacancy loop with $Z_{(c,a)}^{i,2i} = 1.2$.

Appendix E. We select a dose rate of $G_0 = 10^{-7}$ dpa/s at a temperature of 553 K. Then a systematic parametric study was performed, varying the dislocation bias, $Z_{(c,a)}^{i,2i}$, within a range of 1.0–1.5, the diffusion anisotropy factor, p^m , within 0.5–1.0 and the production bias of di-interstitials, f_{2i} , within 0.0–0.5. In Fig. 5, the change in the area of the interstitial and vacancy loop is shown after 0.1 dpa, in square nanometers (nm^2). A constant contour line ($\Delta A = 0$) is shown to distinguish between loop growth and loop shrinkage. A schematic illustration of the flux of defects to the loop segments is shown in Fig. 6 for the conditions labeled 1–4. The ellipticity ratio b/a of the $\langle a \rangle$ loops is shown, where b is the major axis (along $[10\bar{1}0]$) and a is the minor axis (along $[0001]$) of the loops.

Figs. 5(a) and 5(b) show the effects of varying the DAD of di-interstitials, p^m , and varying the dislocation bias, $Z_{(c,a)}^{i,2i}$, on the growth rate of an interstitial $\langle a \rangle$ and vacancy $\langle c \rangle$ loop. Location 1 corresponds to the case of $p^m = 1$ and $Z_{(c,a)}^{i,2i} = 1$, i.e., with isotropic diffusion and without dislocation bias. The vacancy and interstitial loops shrink slowly due to thermal vacancies only; the case discussed in Section 4.1 and shown schematically in Fig. 6(a). Location 2 corresponds to $p^m = 1$ and $Z_{(c,a)}^{i,2i} = 1.5$, where there is no DAD and a high dislocation bias. Under this condition, both the vacancy loop (Fig. 5(b)) and the interstitial loop (Fig. 5(a)) receive a higher flux of interstitials and di-interstitials due to the dislocation bias, leading to the concentric shrinkage of the vacancy loop and growth of the interstitial loop (Fig. 6(c)). Location 4 corresponds to the case of $p^m = 0.5$ and $Z_{(c,a)}^{i,2i} = 1$, i.e., with DAD and without dislocation bias. From Fig. 4, the DAD causes the vacancy concentrations to increase, while the interstitial concentrations decrease compared with the isotropic diffusion case. This creates a large flux of vacancies to the interstitial loop, causing it to shrink. The dislocation loop also absorbs fewer interstitials in the basal plane than along the basal pole, creating a biased net flux depending

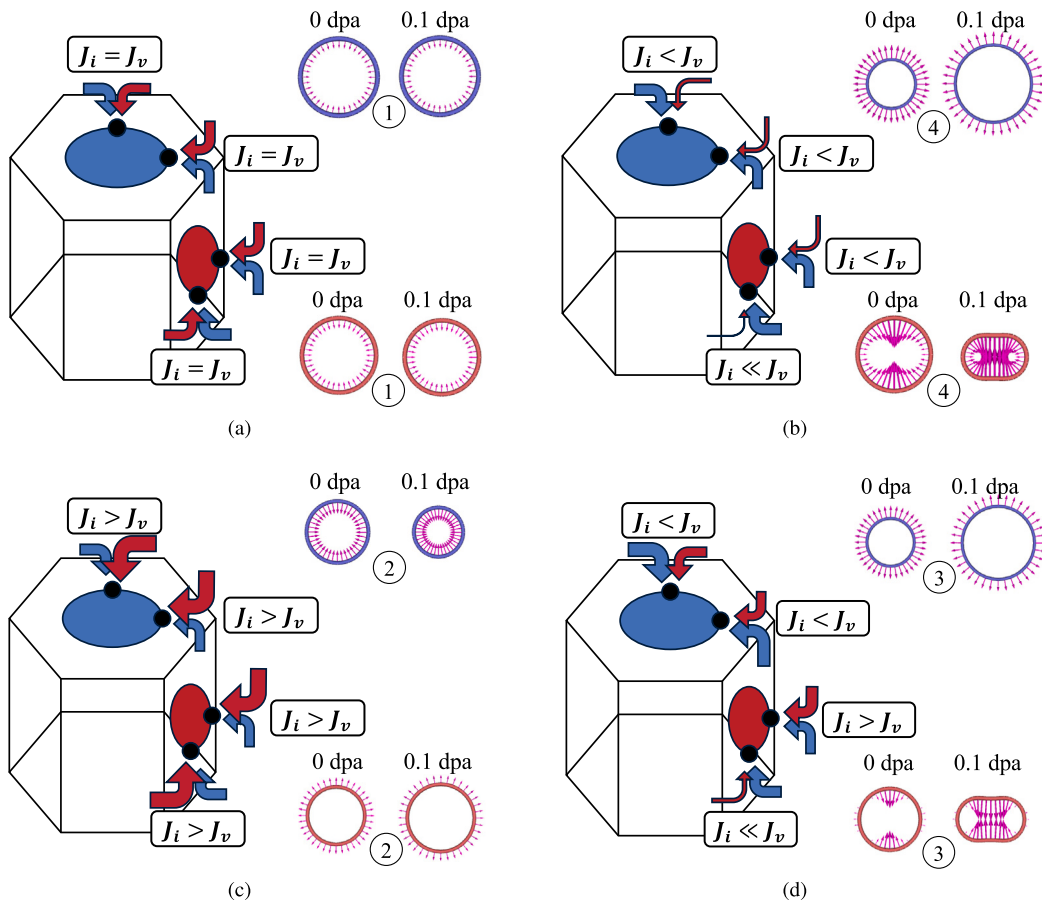


Fig. 6. Schematics of the flux of vacancies and interstitials (i , $2i$ lumped together) to a basal vacancy (blue) and prismatic interstitial (red) loop, from conditions labeled 1–4 in Figs. 5(a) & 5(b). Snapshots are directly from DDD simulations. The dislocation loops and velocities (pink arrows) are scaled for clarity. Interstitial loops are shown to inhabit 1st order prismatic planes for visibility, but inhabit 2nd order prismatic planes in simulations. (a) Isotropic conditions with no dislocation bias. (b) Anisotropic conditions for di-interstitials with no dislocation bias. (c) Isotropic conditions with dislocation bias for interstitial defects ($i, 2i$). (d) Anisotropic conditions for di-interstitials with dislocation bias for interstitial defects ($i, 2i$).

on the segment orientation, and results in an elliptical shape for the prismatic $\langle a \rangle$ loop (Fig. 6(b)). It is noteworthy that this ellipticity of the prismatic loop is a natural outcome of the DAD without parameter adjustment. Moving along the line of constant dislocation bias $Z_{(c,a)}^{i,2i}$ in Fig. 5(a), we see that the higher the diffusion anisotropy difference, p^m , the more elliptical the interstitial loop will become. In contrast, the basal vacancy loop (Fig. 5(b)) exhibits uniform concentric growth at location 4 since both defects diffuse isotopically in the basal plane (Fig. 6(b)). Finally, location 3 corresponds to $p^m = 0.5$ and $Z_{(c,a)}^{i,2i} = 1.5$, with both high DAD and high dislocation bias. The combined effects contribute to asymmetric and elliptical growth of the prismatic interstitial loop, as shown in Fig. 6(d). Compared to location 4, it is clear that the increased dislocation bias mitigates the effect of DAD, resulting in a net interstitial flux ($J_{i,2i} > J_v$) towards dislocation segments with the prismatic orientation, hence the elongation of the loop along $\langle a \rangle$ direction. This clearly shows that the DAD and the elastic bias can contribute oppositely to the $\langle a \rangle$ loop kinetics. The $\langle c \rangle$ vacancy loop grows under these conditions and although the dislocation bias for interstitials is large, the DAD & recombination effects are dominant. Interestingly, phase field modeling by Rouchette et al. (2014) has shown that interstitials are preferentially absorbed by prismatic $\langle a \rangle$ loops rather than their basal $\langle c \rangle$ counterparts due to the anisotropic relaxation volume of interstitials. Thus, a larger dislocation bias of $Z_{(a)}^{i,2i}$ and an intermediate value of DAD are expected to produce the co-growth of prismatic and basal loops.

Figs. 5(c) and 5(d) further present the effects of varying the production bias f_{2i} , the fraction of interstitials directly formed as di-interstitials from the cascade, and varying the DAD of di-interstitials, where the values of p^m are chosen from Table 2. The dislocation bias for interstitials is set as $Z_{(c,a)}^{i,2i} = 1.2$. For the case of $f_{2i} = 0.0$ and $p^m = 1.0$, the prismatic interstitial loop concentrically grows due to the dislocation bias, while the basal vacancy loop shrinks. When increasing the di-interstitials fraction f_{2i} , a larger percentage of defects produced from the cascade diffuse rapidly in the basal plane. Accordingly, there is higher concentrations of vacancies and lower concentrations of interstitials at the center region, thus an interstitial loop shrinks elliptically, but a vacancy loop grows (Fig. 6(b)). Now for fixed p^m with increasing f_{2i} alone, higher f_{2i} induces a larger loop evolution rate, irrespective of

Table 3

Additional model parameters used in simulations involving multiple loops for the irradiation damage production, and properties for di-interstitials. Properties for vacancies and interstitials are found in Table 1. For repeated entries, the values reported here are used instead.

Definition	Symbol	Value	Reference/Comment
Damage Production			
Temperature	T	553 K	Carpenter et al. (1981, 1988)
Dose rate	G_0	10^{-7} dpa/s	Carpenter et al. (1981, 1988)
Fraction of di-interstitials	f_{2i}	0.4	Zhou et al. (2018), Kim et al. (2018)
Mobile Species			
Relative relaxation volume of di-interstitial	$\text{tr}(\chi_i^r)/\Omega_0$	2.4	Vérité et al. (2013)
Formation energy of di-interstitial	E_{2i}^f	6.0 eV	De Diego et al. (2002)
Migration energies of di-interstitial	$E_{2i}^{m(11,22,33)}$	Table 2	
Diffusion coefficient of di-interstitial	D_{2i}^m	8.1882×10^{-10} m ² /s	Li et al. (2020), Christien and Barbu (2009)
Interaction bias factor of vacancies to all loops	$Z_{(c)}^v$	1.0	Woo (1988), Wolfer (2007)
Interaction bias factor of interstitial defects to basal loops	$Z_{(c)}^{i,2i}$	1.05	Woo (1988), Wolfer (2007)
Interaction bias factor of interstitial defects to prismatic loops	$Z_{(a)}^{i,2i}$	Varied	

growth or shrinkage. The biased flux of di-interstitials not only influences the distribution of other defects, but also amplifies the loop kinetics.

4.3. Multiple-loops evolution

The previous studies validate the accuracy of the model and highlight the importance of factors such as defect reactions, diffusion anisotropy difference, dislocation bias, and production bias essential to capture loop kinetics. Nevertheless, irradiation produces dense populations of dislocation loops, where the interactions between loops, their stress fields, and boundary sinks play a critical role in irradiation-induced deformation such as swelling, growth, creep, and hardening. These effects become pronounced when loops are closely spaced, where shielding and stress interactions may reduce defect availability in a dense population of loops. We first analyze the effect of loop spacing on the growth of three basal vacancy loops, as shown in Fig. 7(a). Starting with loops of radius $r_0 = 10$ nm, we systematically vary the distance, d , between the central loop and the outer loops, from 1 to 200 nm, and compare the results with those of a single loop centered in the volume. The size of the edge of the box used for these studies is set to $L = 15465|b|$, where $|b|$ is the magnitude of the Burgers vector. Defect mobilities and parameters for interstitials and vacancies are summarized in Table 1, while the parameters for di-interstitials and bias factors for all species ($v, i, 2i$) are updated in Table 3. Since the bias parameters chosen are $f_{2i} = 0.4$, $Z_{(c)}^{i,2i} = 1.05$, and $p^m = 0.7$, the basal vacancy loops are expected to grow (Fig. 5(b)). In Fig. 7(b), the average loop radius is shown for the three loops, along with the case for a single isolated loop in the center of the simulation volume. As the separation distance d between loops decreases, the overlap of their concentration fields leads to *non-local* competition for mobile point defects. This reduces the local supersaturation near each loop and thus the net defect flux driving loop growth. As a result, the average loop radius at a given dose is significantly reduced when the loops are closely spaced ($d/r_0 < 5$) compared to isolated growth conditions. This demonstrates a key *non-local* effect in spatially resolved dislocation climb kinetics (Gu et al., 2015). At very large spacings, the *non-local* effect nearly vanishes, and the loop growth rate begins to converge to the single loop case.

We next focus on simulating the evolution of a population of dislocation loops in irradiated and unstressed zirconium consistent with the microstructure evolution of the irradiation-induced growth (IIG) phenomenon. IIG is characterized by a net expansion along the $\langle a \rangle$ axis and a contraction along the $\langle c \rangle$ axis of the crystal, associated with the agglomeration of irradiation-induced defect clusters and the simultaneous growth of the basal $\langle c \rangle$ vacancy and the prismatic $\langle a \rangle$ interstitial loops (Adamson et al., 2019). Experiments with annealed and irradiated Zr have shown three stages of growth (Carpenter et al., 1988): (I) an initial transient, (II) a saturation of growth related with the formation of clusters and $\langle a \rangle$ dislocation loops (Bacon et al., 2003; Voskoboinikov et al., 2006), and (III) breakaway growth after ~ 3 dpa, corresponding to the formation and growth of the basal vacancy dislocation loops. To reproduce stage III of the IIG, we consider point defects, di-interstitials, vacancy $\langle c \rangle$ dislocation loops, and interstitial $\langle a \rangle$ dislocation loops at a temperature of 553 K. It should be pointed out that TEM observations have revealed the coexistence of vacancy and interstitial $\langle a \rangle$ loops (Jostsons et al., 1977), and the co-growth of these prismatic loops is still debated (March-Rico et al., 2025). However, given the current simulation temperature of 553 K, and focus of stage III, we include only interstitial $\langle a \rangle$ dislocation loops hereafter.⁵ Typical densities for $\langle a \rangle$ interstitial loops at stage III are 10^{20} – 10^{22} $\mathcal{L}/\text{m}^3$ (Gilbert et al., 1979; Kobylansky et al., 2008), and for $\langle c \rangle$ vacancy loops are 10^{20} – 10^{21} $\mathcal{L}/\text{m}^3$ (Tournadre et al., 2012). The initial microstructures generated for simulations consist of total loop densities of 2.5 and 5×10^{20} $\mathcal{L}/\text{m}^3$, where 90% are $\langle a \rangle$ interstitial loops ($r = 6$ nm), and the remaining are $\langle c \rangle$ vacancy loops ($r = 14$ nm). This initial microstructure produces swelling of the same magnitude as growth strains, as reported in Carpenter et al. (1988). The simulation parameters are shown in Tables 1 and 3, where all repeated parameters are taken from the latter. It should

⁵ TEM experiments showed that at 623 K, 10% of $\langle a \rangle$ loops are of vacancy type, and this decreases with decreasing temperature (Liu et al., 2020). Furthermore, the majority of vacancy $\langle a \rangle$ loops have been shown to shrink during stage II, and $\sim 80\%$ of $\langle a \rangle$ loops are interstitial (Griffiths, 1988). The model is capable of capturing the collective behavior of vacancy and interstitial $\langle a \rangle$ loops, and this will be presented in a separate work.

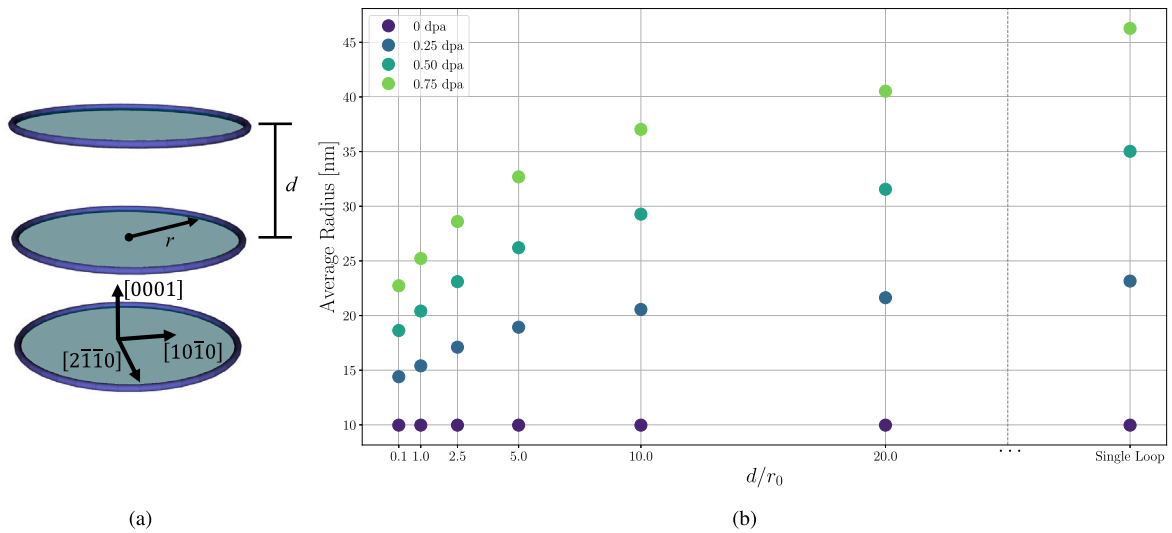


Fig. 7. Non-local effects of dislocation loops on their growth rate. (a) Schematic of simulation set up of three basal vacancy dislocation loops with an initial radius $r_0 = 10$ nm, placed on the basal plane of a simulation domain of size $L = 5$ μm , and separated by a distance d . (b) Loop radius evolution up to 0.75 dpa for the average loop radius for varying separation distances d/r_0 where r_0 is the initial loop radius. A single centered vacancy loop is shown for comparison.

be noted that the objective of the present work is to make qualitative observations on the microstructure evolution of irradiated Zr but not to reproduce the experimentally observed growth strains. A detailed investigation of the IIG can be found in our recent paper (Maron et al., 2024).

Fig. 8(a) presents the total plastic strain along $\langle a \rangle$ and $\langle c \rangle$ axes as a function of irradiation dose in a cubic domain with $L = 1$ μm side length. The DAD of di-interstitials, p^m , was varied from 0.6–0.8 following Table 2, and the interaction bias, $Z_{(a)}^{i,2i}$ was varied from 1.1–1.3. The latter condition of $Z_{(a)}^{i,2i} > Z_{(c)}^{i,2i}$ is supported by the works of V  rit   et al. (2013) and Rouchette et al. (2014), which is a result of the anisotropic elastic interactions between interstitials and prismatic/basal loops. Globally, it is obvious that DAD significantly influences the overall growth strain, where larger p^m increases the growth rate of basal $\langle c \rangle$ loops but decreases the rate of prismatic $\langle a \rangle$ loops. It is important to note that the loop kinetics is highly correlated with position. As di-interstitials diffuse anisotropically, a large amount of interstitials are accumulated at the top and bottom region of the crystal, owing to its small diffusivity along $\langle c \rangle$ direction (see also the concentration profiles in Fig. 4). This leads to a preferential growth of interstitial type loops in this region, while the vacancy type loops tend to grow at the center region of the crystal with high local vacancy concentrations. This so-called anti-correlation of vacancy and interstitial type loops is fully consistent with simulation results of the spatially resolved cluster dynamics (Maron et al., 2024), as well as recent TEM observations (Long et al., 2024). With a sufficiently small DAD $p^m \leq 0.6$, interstitial concentrations are dramatically reduced, suppressing growth strains in the $\langle a \rangle$ directions. Only a very small region of the crystal provides favorable conditions for interstitial loop growth, as shown in the single loop diagram Fig. 4(d). Meanwhile, vacancy concentrations increase significantly, promoting the growth of nearly all $\langle c \rangle$ loops and inducing a large strain along the $\langle c \rangle$ direction. Alternatively, with a weak DAD of $p^m = 0.8$, an opposite trend is observed, where the growth regions of the interstitial loops become larger and the vacancy concentrations are reduced. This induces a decrease of the strain along the basal pole and is in contrast to the IIG phenomenon. We further find that the dislocation bias $Z_{(a)}^{i,2i}$ has little impact on the observed trends, but can strongly enhance the growth rate of both prismatic and basal loops.

The dislocation loop size distributions of a selected simulation with $p^m = 0.7$ and $Z_{(a)}^{i,2i} = 1.2$ are shown in Figs. 8(b)–8(e), and correspond to the dashed growth strain curve in Fig. 8(a). Here, each individual dislocation loop is presented as a circular marker characterized by its size and position along the basal pole. We find that $\langle c \rangle$ vacancy loops tend to grow in the central regions of the crystal (Fig. 8(c)), whereas interstitial $\langle a \rangle$ loops generally grow near the top and bottom of the basal pole (Fig. 8(d)). This anti-correlation is more pronounced with increasing dose, where many of the $\langle a \rangle$ loops in the central region are fully annihilated after 2 dpa. This is further supported by the ellipticity distribution of Fig. 8(e), where the highly elliptical $\langle a \rangle$ loops found near the center of the simulation volume are shrinking. The representative dislocation loop geometries at different positions are shown as the insets in Fig. 8(e). Inset 1 shows the growing loop near the top of the crystal, meanwhile inset 2 presents the highly elliptical loop that shrinks near the center of the crystal. Inset $\star$ also shows an irregularly shaped loop, which is found near a boundary parallel to the basal pole. Irregular-shaped loops have also been observed in experiments (Long et al., 2024), whose formation, we believe, is related to the boundary effect that causes a strong defect concentration gradient.

Now introducing a higher total loop density of 5×10^{20} L/m^3 , from Fig. 9, it is seen that the effects of the DAD and dislocation bias remain consistent with the lower density case. Specifically, for $p^m = 0.6$, highly anisotropic diffusion suppresses interstitial loop growth, while elevated vacancy concentrations promote growth of nearly all vacancy loops. In contrast, for $p^m = 0.8$, the growth of

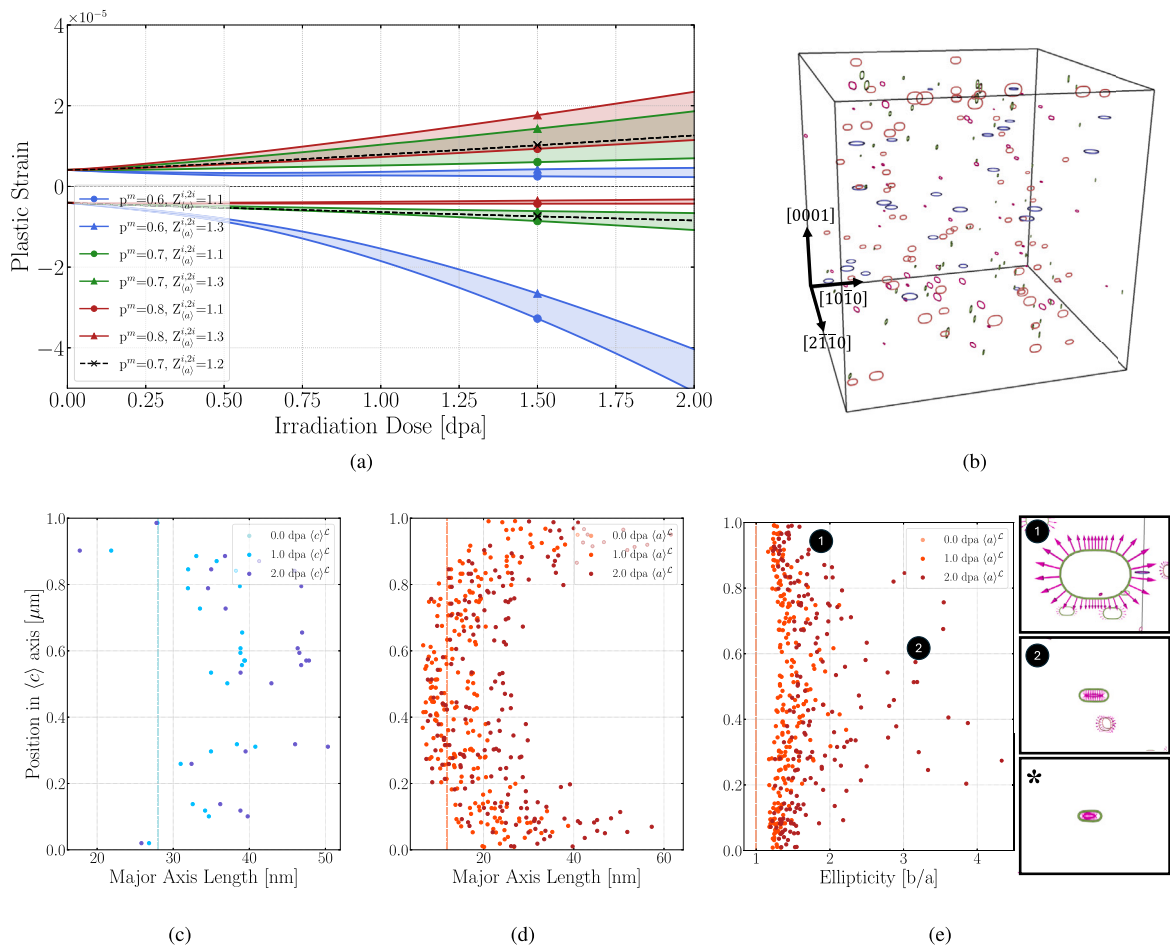


Fig. 8. Simulation results for a dislocation loop density of $2.5 \times 10^{20} \text{ L/m}^3$ (90% $\langle a \rangle$ interstitial and 10% $\langle c \rangle$ vacancy loops) in a simulation domain of side length $L = 1 \mu\text{m}$. (a) Total plastic strains $\frac{1}{2}(\epsilon_{11} + \epsilon_{22})$ and ϵ_{33} up to 2 dpa. (b) Snapshot of simulation domain at 2 dpa. (c) Vacancy $\langle c \rangle$ loop major axis length vs. position along the $\langle c \rangle$ axis. (d) Interstitial $\langle a \rangle$ loop major axis length vs. position along the $\langle c \rangle$ axis. (e) Interstitial $\langle a \rangle$ loop ellipticity vs. position along the $\langle c \rangle$ axis. Insets are snapshots of dislocation loops at different regions in the crystal.

the vacancy loops is limited. In the intermediate regime with $p^m = 0.7$, a co-growth is observed, where interstitial loops grow closer towards boundaries perpendicular to the basal pole, while vacancy loops grow in the center region of the crystal, consistent with the earlier observations. The corresponding vacancy concentration profiles are shown in Fig. 10, for doses of 0 dpa (Fig. 10(a)) and 2 dpa (Fig. 10(b)). It is clear that the high concentration of vacancies is found near the boundaries parallel to the [0001] axis due to the DAD and vacancy-interstitial interactions. In addition, the vacancy concentration is strongly influenced by the dislocation loops, much lower in regions with a high density of loops. At 2 dpa, the total concentration of vacancies in these regions has decreased, since as the loops grow, their sink strength grows. The vacancy (and i, 2i) concentration profiles create regions of preferential $\langle a \rangle$ and $\langle c \rangle$ loop growth. This is similarly pronounced for different initial densities of 2.5 or $5 \times 10^{20} \text{ L/m}^3$ highlighting the dominant effect of DAD. However, at higher density, the ellipticity of interstitial loops does not change as significantly with increasing doses (Fig. 9(e)). The increased density of defects enhances the *non-local* effect, naturally leading to slower evolution kinetics. The insets 1–3 of Fig. 9(e) show individual interstitial loops from different locations along the $\langle c \rangle$ axis of the crystal. In the center of the crystal the interstitial loops shrink highly elliptically, as their climb velocities are larger in the $\langle c \rangle$ -direction than in the $\langle a \rangle$ -directions. At the top and bottom of the basal pole, however, from inset 1, the DAD and defect interactions lead to interstitial loop growth, with slightly higher climb velocity in the $\langle a \rangle$ -direction than in the $\langle c \rangle$ -direction, resulting in a less pronounced ellipticity. Loops with larger ellipticity exhibit stronger asymmetric nodal velocities, making them even more elliptical and affecting subsequent loop evolution.

5. Discussion

From the variations of DAD, the dislocation bias, and the defect densities studied here, we suggest that not all values of p^m lead to growth along both the $\langle a \rangle$ and $\langle c \rangle$ axes. This contrasts with cluster dynamics models, where all values of the anisotropy factor

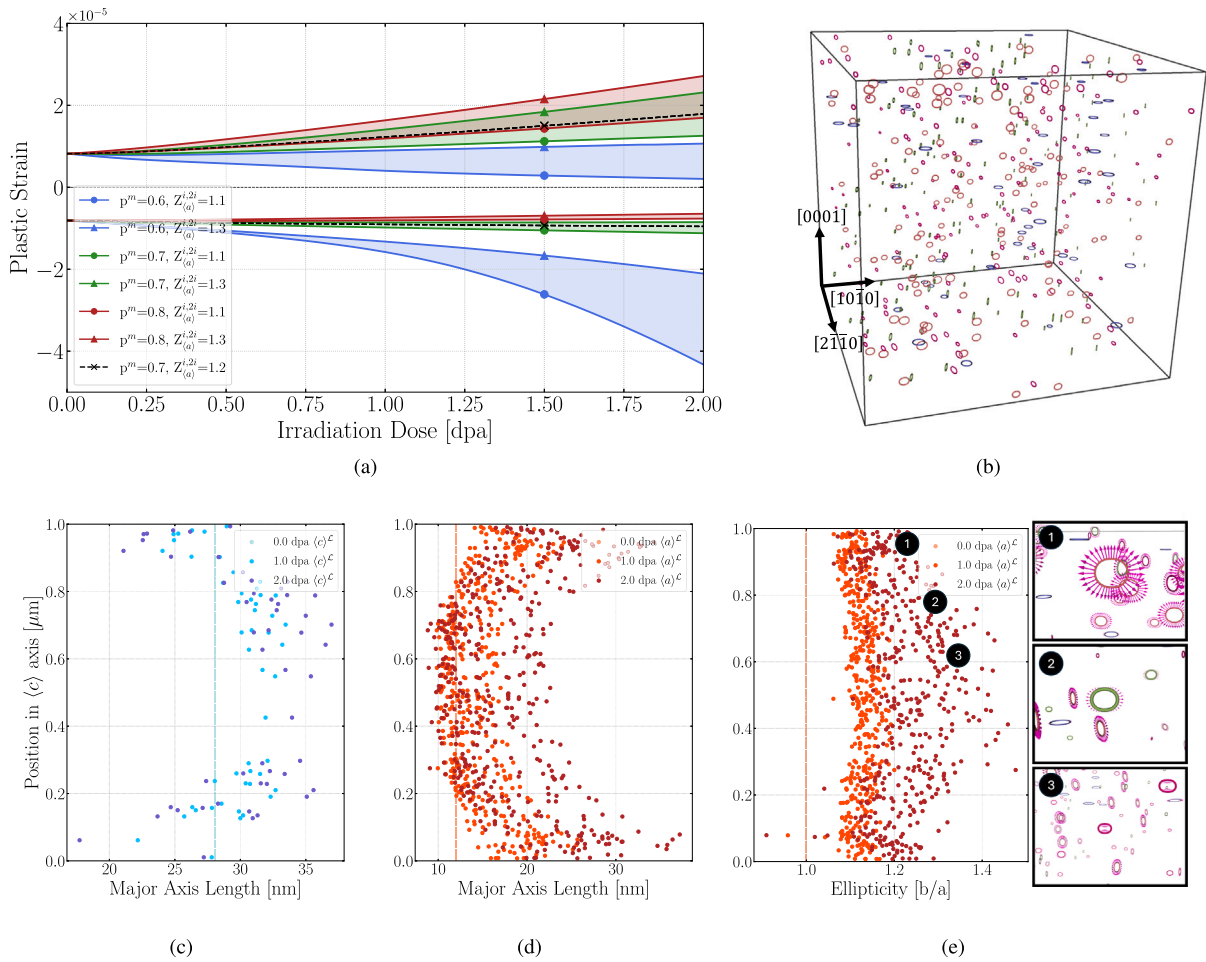


Fig. 9. Simulation results for a dislocation loop density of $5 \times 10^{20} \text{ L/m}^3$ (90% $\langle a \rangle$ interstitial and 10% $\langle c \rangle$ vacancy loops) in a simulation domain of side length $L = 1 \mu\text{m}$. (a) Total plastic strains $\frac{1}{2}(\epsilon_{11} + \epsilon_{22})$ and ϵ_{33} up to 2 dpa. (b) Snapshot of simulation domain at 2 dpa. (c) Vacancy $\langle c \rangle$ loop major axis length vs. position along the $\langle c \rangle$ axis. (d) Interstitial $\langle a \rangle$ loop major axis length vs. position along the $\langle c \rangle$ axis. (e) Interstitial $\langle a \rangle$ loop ellipticity vs. position along the $\langle c \rangle$ axis. Insets are of snapshots of dislocation loops at different regions in the crystal.

($p^m < 1$) promote greater loop growth, except under isotropic conditions where $p^m = 1$ (Maron et al., 2024; Christien and Barbu, 2009; Zhang et al., 2023). The results presented here show that at low anisotropy, $0.8 < p^m < 1.0$, the growth of the vacancy loop is restricted due to higher interstitial concentrations, leading to the growth of interstitial prismatic loops only. However, excessive anisotropy can restrict the regions available for interstitial loop growth, and only basal vacancy loops grow. It is therefore reasonable to assume that only a specific range of the anisotropy factor permits co-growth of $\langle a \rangle$ and $\langle c \rangle$ loops, and that values of $p^m > 0.8$ are unreasonable. Indeed, early MD simulations showed that the anisotropy factor $p^m \approx 0.75$ for interstitials (Woo and Liu, 2007; Osetsy et al., 2002). More recent atomistic simulations have found $p^m \approx 0.85$ for interstitials and $p^m \approx 0.5$ for di-interstitials (March-Rico and Wirth, 2022; Sakaël et al., 2025). As shown in Appendix E, simulation volume can impact loop growth kinetics, particularly for smaller crystals/grains, therefore, this may deviate at smaller crystal sizes. Another observation drawn from the collective loop growth is that the overall growth rates of all loops are reduced as the loop density increases. Although long-range elastic interactions between loops can influence defect fluxes by perturbing equilibrium concentrations, for the densities and temperature studied here (2.5 & $5 \times 10^{25} \text{ L/m}^3$ at 553 K), the dominant mechanism responsible for the reduced loop growth rate is the increase in effective sink strength and *non-local* effects. With increasing dislocation density, elastic interactions between dislocations may become the dominant mechanism. In addition, loops that become elliptical tend to shrink/grow much faster (Figs. 8(e) and 9(e)), and the orientation of segments along a dislocation loop may lead to opposite behavior from the DAD (Fig. 6(d)), further highlighting the importance in incorporating full degrees of freedom of dislocation loops within DDD. At high doses and elevated temperatures ($T > 600 \text{ K}$), Zr has been shown to form networks, and the ellipticity of $\langle a \rangle$ loops seems to be a key mechanism promoting its formation (Warwick et al., 2021). This phenomenon would be interesting to study further using the developed framework.

TEM observations of the evolution of dislocation loops in irradiated Zr have consistently shown that the $\langle a \rangle$ dislocation loops are elliptical, with the minor axis of the loop in the (0001) plane and the major axis in the vicinity of the [0001] direction. The

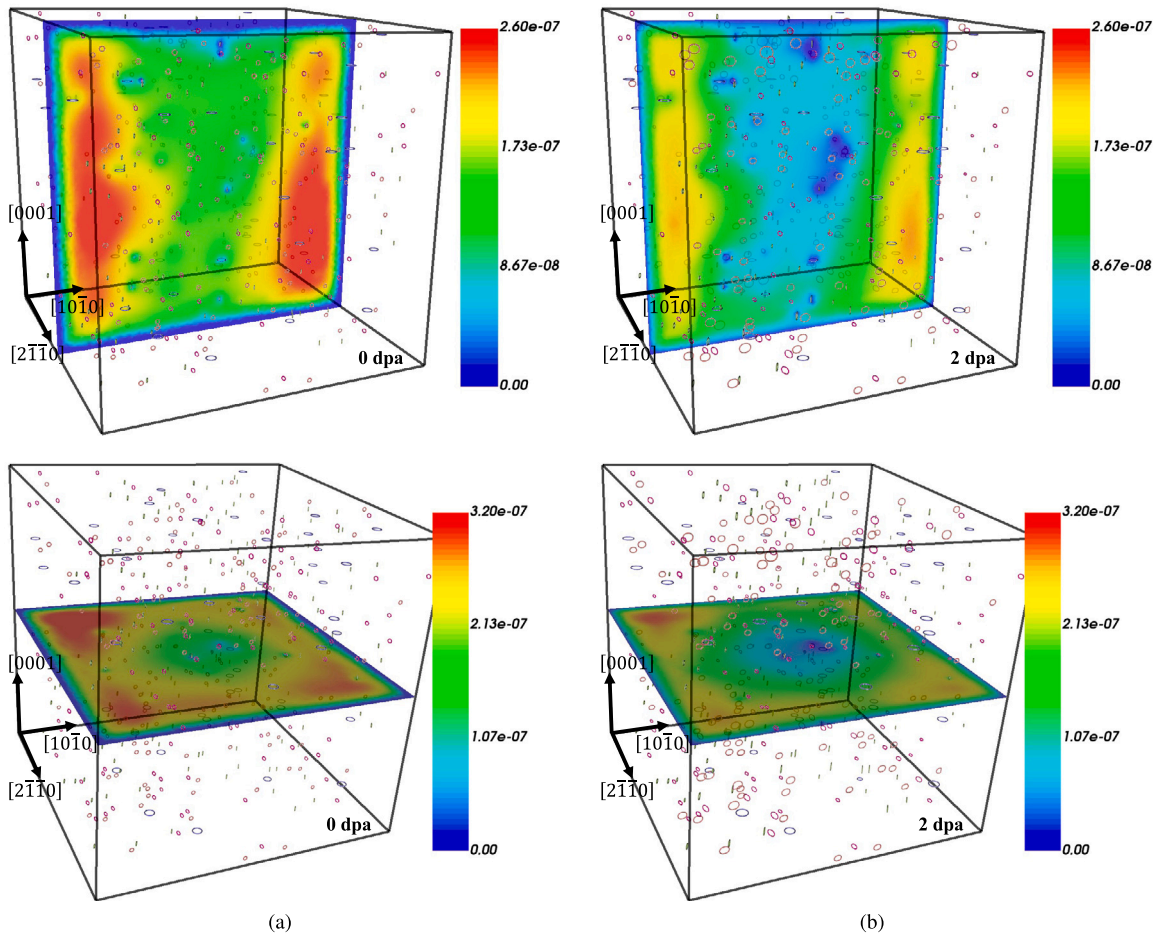


Fig. 10. Vacancy concentration field plots of a plane perpendicular to the $\langle a_1 \rangle$ -axis (top) and perpendicular to the $\langle c \rangle$ -axis (bottom) for a dislocation density of $5 \times 10^{20} \text{ L/m}^3$, $p^m = 0.7$, and $Z_{(a)}^{i,2i} = 1.2$. (a) Vacancy concentration fields at 0 dpa. (b) Vacancy concentration fields at 2 dpa.

ellipticity was particularly marked for vacancy $\langle a \rangle$ loops, whereas interstitial $\langle a \rangle$ loops tended to be nearly circular (Jostons et al., 1977; Gilbert et al., 1979; Northwood et al., 1979). In this work the ellipticity of the interstitial $\langle a \rangle$ loops was such that the major axis was in the (0001) plane and minor axis in the [0001] direction resulting in $b/a > 1$, where b is the major axis of the loop. The interstitial $\langle a \rangle$ loops began circular and anisotropic elasticity was not considered; however, if the anisotropic elasticity of Zr was considered, the loops would tend to reduce their ellipticity and become near-circular, consistent with the TEM observations. Furthermore, this effect would be the opposite for the vacancy $\langle a \rangle$ loops, and they would retain their ellipticity, which has also been observed via TEM (Liu and Han, 2021). This supports the predictions of Woo (1988) that initially $\langle a \rangle$ loops have major axis in the [0001]-direction in elastically anisotropic materials. We confirm here that variations in dislocation loop ellipticities are caused by DAD, where the absorption efficiency of dislocation segments is a function of their line direction, giving rise to a bias differential among different segments of the same dislocation loop. An $\langle a \rangle$ loop segment perpendicular to the basal pole will generally receive a higher flux of interstitials than segments perpendicular to the basal plane, which in conjunction with the elastic anisotropy produce the elliptical vacancy $\langle a \rangle$ loops and near-circular interstitial $\langle a \rangle$ loops observed experimentally. It should also be noted that the pipe diffusion is not considered in the present work, which could further influence the shape of dislocation loops, particularly at low temperatures.

Another critical observation made in these studies is that loop kinetics and ellipticity are highly dependent on spatial location in the crystal. Loop growth is non-uniform, with vacancy loops preferentially growing in the central regions of the crystal and close to boundaries parallel to the basal pole (Fig. 9(c)). Meanwhile, interstitial loops grow near the top and bottom of the basal pole (Fig. 9(d)). This matches well with the loop density profile quantified from irradiated Zr foils (Fig. 2 in Buckley et al. (1980)), and recent TEM observations of vacancy loops showing similar grain boundary effects (Long et al., 2024). The experimental TEM work by Long et al. (2024) also shows evidence of irregularly shaped $\langle a \rangle$ loops, which we find here (Fig. 8(d)-*) to be due to concentration gradients of defects near grain boundaries. Other experimental work has shown abnormal growth of loops near dislocation lines (Li

et al., 2024a), which act as strong sinks for irradiation-induced defects. Thus, in regions where local defect flux is greatly affected by the *non-local* effects of dislocation lines, high density of dislocation loops, or grain boundaries, abnormally-shaped loops may be found. The spatial information of dislocation loops is thus necessary, and could explain TEM observations where loop character, size, and ellipticities varied significantly from grain to grain (Jostsons et al., 1977; Griffiths, 1988). The spatial information within a crystal or grain should be taken into account when performing experimental TEM on the dislocation loop size and shape distributions within irradiated samples. The spatial heterogeneity in loop sizes and shapes also underscores the critical role of incorporating both anisotropic diffusion and elastic interactions in models for irradiation damage.

Finally, we briefly address the absence of aligned prismatic $\langle a \rangle$ loops, often referred to as corduroy patterns, in our results. Although such an alignment has been observed in irradiated Zr experimentally (Kelly et al., 1976; Gilbert et al., 1979; Northwood et al., 1979), the origin of this phenomenon remains unresolved. Several mechanisms have been proposed, including pipe diffusion (Breidi and Dudarev, 2022), transformation kinetics of vacancy clusters (Sakaël et al., 2023), and glide-mediated self-organization (Cui et al., 2022), but none have been definitively validated. Notably, recent TEM observations and MD simulations suggest that many prismatic $\langle a \rangle$ loops are randomly distributed within basal planes (Dai et al., 2019), consistent with our findings and reinforcing that anisotropic diffusion alone does not lead to corduroy formation. A more comprehensive investigation that incorporates additional mechanisms such as loop mobility, transformation kinetics, and pipe diffusion is needed to resolve this phenomenon and can be the topic of future work upon expanding the current framework.

6. Conclusions

We have successfully developed a novel 3D discrete dislocation dynamics framework coupled to cluster dynamics to investigate defect reaction–diffusion and dislocation climb in finite crystals subject to radiation conditions, incorporating the diffusion anisotropy difference, the dislocation bias, and the production bias. Overall, the findings here provide a comprehensive framework for understanding the complex mechanisms governing loop growth under irradiation and offer valuable insights to refine theoretical models and simulations. The results emphasize that accurate predictions of dislocation behavior require careful calibration of defect mobility parameters and an integrated approach that accounts for anisotropic diffusion, dislocation bias, elastic interactions, and three-dimensional spatial resolution in determining irradiation-induced defect distributions. The main conclusions can be summarized as follows.

1. The diffusion anisotropy difference (DAD) has a much larger effect on the kinetics of individual dislocation loop evolution than the dislocation bias ($Z_{\langle c,a \rangle}^{i,2i}$) and the cascade production bias ($f_{i,2i}$).
2. With elastic isotropy, circular interstitial $\langle a \rangle$ dislocation loops will grow or shrink elliptically with major axis in the basal plane and minor axis in the basal pole. The additional degrees of freedom have a drastic influence on the loop kinetics, leading to a significant deviation from the circular approximation.
3. Due to the effects of defect reactions, diffusion anisotropy difference, and boundary effects, there is a spatial dependence of loop behavior. Basal vacancy loops grow in regions near the crystal center in close proximity to boundaries parallel to the basal pole. Meanwhile, prismatic interstitial loops grow in regions in close proximity to boundaries perpendicular to the basal pole, at the top and bottom of a crystal. Experimental analysis of irradiated samples should take into account the location of the sample being observed, as there is a dependence of size distribution, loop character, and ellipticity on location within the crystal.
4. In general, a higher diffusional anisotropy will not always result in higher growth strains along both the $\langle a \rangle$ and $\langle c \rangle$ axes simultaneously when considering the spatial resolution. p^m values to describe the anisotropy of interstitial type defects should be less than 0.8 to produce simultaneous $\langle a \rangle$ axis expansion and $\langle c \rangle$ axis contraction. Considering loop densities approaching TEM observations, a higher anisotropy (lower values of p^m) is likely more appropriate.
5. The experimentally observed spatial separation between basal vacancy loops and prismatic interstitial loops can be explained through the effects of DAD, defect reactions, and boundary effects. In many regions where interstitial $\langle a \rangle$ loops grow, vacancy $\langle c \rangle$ loops generally cannot, and vice versa.

CRediT authorship contribution statement

Yang Li: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Methodology, Funding acquisition, Formal analysis, Conceptualization. **Matthew Maron:** Writing – review & editing, Writing – original draft, Validation, Supervision, Software, Methodology, Formal analysis, Data curation, Conceptualization. **Kristopher Baker:** Writing – review & editing, Supervision, Funding acquisition. **Benjamin Ramirez Flores:** Writing – review & editing, Supervision, Funding acquisition. **Thomas Black:** Writing – review & editing, Supervision, Funding acquisition. **James Hollenbeck:** Writing – review & editing, Supervision, Funding acquisition. **Inam Lalani:** Visualization, Software, Data curation. **Nasr Ghoniem:** Writing – review & editing, Supervision, Project administration, Methodology, Investigation, Conceptualization. **Giacomo Po:** Writing – review & editing, Supervision, Software, Project administration, Investigation, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

Y.L. acknowledges the support of the National Natural Science Foundation of China under Grant No. 12302279. The support from the Naval Nuclear Laboratory (NNL) under PO#126180 in collaboration with Multiscale Materials Solutions (MMS), LLC is also acknowledged. N.G. acknowledges the support by the US Department of Energy, Office of Fusion Energy Sciences, under Award Number DE-SC0018410 in partnership with UCLA. G.P. and M.M. acknowledge the support of the US Department of Energy, Office of Fusion Energy Sciences, under Awards Numbers DE-SC0024675 and DE-SC0024401 with the University of Miami, and the US National Science Foundation through the CAREER grant number 2340174 with the University of Miami.

Appendix A. Analytical expression of c^∞ generated by a straight dislocation segment

Suppose that the loop $\mathcal{L}$ in Eq. (48) is discretized into straight segments, then the loop integral can be expressed as the sum integrals over straight segments. The contribution to c^∞ of a generic segment connecting node A' to node B' is

$$c_{A'B'}^\infty(\mathbf{x}) = \int_{A'}^{B'} \frac{(\mathbf{b}' \times \boldsymbol{\xi}') \cdot \mathbf{w}_\star'}{4\pi \sqrt{\det(\mathbf{D}_\star)(\mathbf{x} - \mathbf{x}') \cdot \mathbf{D}_\star^{-1}(\mathbf{x} - \mathbf{x}')}} d\ell' = \frac{L_{A'B'}}{4\pi} \int_0^1 \frac{(\mathbf{b}' \times \boldsymbol{\xi}') \cdot \mathbf{w}_\star'(u')}{\sqrt{\det(\mathbf{D}_\star)(\mathbf{x} - \mathbf{x}'(u')) \cdot \mathbf{D}_\star^{-1}(\mathbf{x} - \mathbf{x}'(u'))}} du'. \quad (\text{A.1})$$

where u' is a parameter ranging from 0 to 1 along the segment. For an “isoparametric segment” the velocity $\mathbf{w}_\star'(u')$ and position $\mathbf{x}'(u')$ are interpolated along the segment using the same shape functions. Assuming linear shape functions, we thus have

$$\mathbf{x}'(u') = \mathbf{x}_{A'}(1 - u') + \mathbf{x}_{B'}u' \quad \mathbf{w}_\star'(u') = \mathbf{w}_{\star A'}(1 - u') + \mathbf{w}_{\star B'}u'. \quad (\text{A.2})$$

Then it yields

$$c_{A'B'}^\infty(\mathbf{x}) = \frac{L_{A'B'}}{4\pi \sqrt{\det(\mathbf{D}_\star)}} \int_0^1 \frac{[1 - u', u']}{\sqrt{au^2 + bu + c}} du \begin{bmatrix} (\mathbf{b}' \times \boldsymbol{\xi}') \cdot \mathbf{w}_{\star A'} \\ (\mathbf{b}' \times \boldsymbol{\xi}') \cdot \mathbf{w}_{\star B'} \end{bmatrix}. \quad (\text{A.3})$$

where

$$a = (\mathbf{x}_{B'} - \mathbf{x}_{A'}) \cdot \mathbf{D}_\star^{-1}(\mathbf{x}_{B'} - \mathbf{x}_{A'}) \quad (\text{A.4})$$

$$b = -(\mathbf{x}_{B'} - \mathbf{x}_{A'}) \cdot \mathbf{D}_\star^{-1}(\mathbf{x} - \mathbf{x}_{A'}) - (\mathbf{x} - \mathbf{x}_{A'}) \cdot \mathbf{D}_\star^{-1}(\mathbf{x}_{B'} - \mathbf{x}_{A'}) \quad (\text{A.5})$$

$$c = (\mathbf{x} - \mathbf{x}_{A'}) \cdot \mathbf{D}_\star^{-1}(\mathbf{x} - \mathbf{x}_{A'}) + \epsilon^2 \quad (\text{A.6})$$

Here ϵ^2 is a small number that has been added to regularize the integral. These integrals can be computed analytically. In particular we have

$$I_{A'}(\mathbf{x}) = \sqrt{a} \int_0^1 \frac{1 - u}{\sqrt{au^2 + bu + c}} du = -\sqrt{1 + \frac{b}{a} + \frac{c}{a}} + \sqrt{\frac{c}{a}} + (1 + \frac{b}{2a}) \ln \frac{2\sqrt{1 + \frac{b}{a} + \frac{c}{a}} + 2 + \frac{b}{a}}{2\sqrt{\frac{c}{a} + \frac{b}{a}}} \quad (\text{A.7})$$

$$I_{B'}(\mathbf{x}) = \sqrt{a} \int_0^1 \frac{u}{\sqrt{au^2 + bu + c}} du = \sqrt{1 + \frac{b}{a} + \frac{c}{a}} - \sqrt{\frac{c}{a}} - \frac{b}{2a} \ln \frac{2\sqrt{1 + \frac{b}{a} + \frac{c}{a}} + 2 + \frac{b}{a}}{2\sqrt{\frac{c}{a} + \frac{b}{a}}} \quad (\text{A.8})$$

Finally $c_{A'B'}^\infty(\mathbf{x})$ becomes

$$c_{A'B'}^\infty(\mathbf{x}) = \frac{L_{A'B'}}{4\pi \sqrt{\det(\mathbf{D}_\star)} \sqrt{a}} [I_{A'}(\mathbf{x}), I_{B'}(\mathbf{x})] \begin{bmatrix} (\mathbf{b}' \times \boldsymbol{\xi}') \cdot \mathbf{w}_{\star A'} \\ (\mathbf{b}' \times \boldsymbol{\xi}') \cdot \mathbf{w}_{\star B'} \end{bmatrix}. \quad (\text{A.9})$$

Appendix B. Vacancy diffusion coefficient in Zr

The vacancy diffusion coefficient in Zr is still unclear, where the value presents a many orders of magnitude difference between experimental measurements and atomistic-scale estimations, as summarized in Table B.1. It should be pointed out that the value measured from experiments consists of the self-diffusion coefficient D_s , which yields

$$D_s = D_0 \exp\left(-\frac{E_f + E_m}{k_B T}\right) \quad (\text{B.1})$$

where D_0 is the pre-exponential factor and $E_f + E_m$ is the activation energy, i.e., the sum of the monovacancy formation energy and the migration energy.

Given that the vacancy formation energy for Zr is of ~ 1.4 – 1.9 eV (Ouyang et al., 2018), the migration energy estimated from experiments is in general higher than that computed from the atomistic simulations, together with a much lower pre-exponential factor, giving rise to a significantly smaller diffusion coefficient for a fixed temperature. This discrepancy is well-known, as simulations study the diffusion of a single vacancy within an ideal crystal. In contrast, solute atoms in real experiments can easily trap vacancies, resulting in a much smaller diffusion coefficient of vacancies compared to simulations. Moreover, Perez et al. (2003) showed that the diffusion coefficient of monovacancies exhibits distinct behaviors at low (< 900 K) and high (> 900 K) temperatures,

Table B.1

Vacancy diffusion coefficients from available literature.

Method	D_0 (m ² /s)			E_m (eV)			Reference
	D_0^{total}	D_0^{basal}	D_0^c	E_m^{total}	E_m^{basal}	E_m^c	
Experiments		2.1×10^{-11}		1.17 (activation energy)			Dymant and Libanati (1968)
Experiments		1.84×10^{-2}		3.31 (activation energy)			Hood (1988)
–		$1.1/6 \times 10^{-13}$		3.48/1.3 (activation energy)			Perez et al. (2003)
Estimated from Horvath et al. (1984)		4.9638×10^{-10}			0.9		Christien and Barbu (2009), Maron et al. (2024)
MD (Ackland1995)	22×10^{-7}	–	–	0.93	0.91	0.96	Osetsy et al. (2002)
MD (PM1998)	–	–	7.19×10^{-9}	–	0.51	0.53	Ruiz et al. (2005)
MD (BMD19)	1.8×10^{-6}	1.6×10^{-6}	2.2×10^{-6}	0.61	0.59	0.67	March-Rico and Wirth (2022)

inducing two different sets of parameters in Zr. The limitation of the simulation temperature may therefore lead to deviations. On the other hand, the diffusion coefficient obtained from molecular dynamics simulations strongly depends on the atomic potentials. For instance, March-Rico and Wirth (2022) have shown that the vacancy is even immobile by adopting the M07 potential (Mendelev and Ackland, 2007). The migration energy obtained from most MD simulations falls within a range of ~ 0.5 – 0.9 eV, and is consistent with the molecular static simulations, but the pre-exponential factor presents a significant difference. In conclusion, the monovacancies diffuse quasi-isotropically in Zr, but their exact diffusion coefficient is still undetermined. In the present work, we adopt the relatively small diffusion coefficient that is estimated from experimental measurements.

Appendix C. Reference case considering point defects only

If we assume that only point defects are generated from the radiation cascade and that the sole type of sinks to point defects is dislocations, the point defect concentrations are simply:

$$\frac{1}{J\Omega_0} \dot{c}_v = -\nabla \cdot \mathbf{J}_v - \alpha_{iv} c_v c_i + p_v + G \quad (C.1)$$

$$\frac{1}{J\Omega_0} \dot{c}_i = -\nabla \cdot \mathbf{J}_i - \alpha_{iv} c_v c_i + p_i + G \quad (C.2)$$

where p_v and p_i are flux of vacancies or interstitials towards the dislocation sinks, α_{iv} the vacancy-interstitial recombination rate and G is the point defect generation rate (per atomic volume).

We can further assume that $\dot{c} = 0$ at the steady state and that both interstitials and vacancies perform isotropic diffusion, the above equations can be written as

$$0 = \frac{D_v}{\Omega_0} \Delta c_v - \alpha_{iv} c_v c_i + p_v + G \quad (C.3)$$

$$0 = \frac{D_i}{\Omega_0} \Delta c_i - \alpha_{iv} c_v c_i + p_i + G \quad (C.4)$$

If we omit the vacancy-interstitial recombination and distinguish the climb velocity of dislocations resulted from the absorption of vacancies or interstitials, we then have

$$D_v \Delta c_v = -p_v - \Omega G \quad (C.5)$$

$$D_i \Delta c_i = -p_i - \Omega G \quad (C.6)$$

Applying the Green's formula G_f , the vacancy and interstitial concentrations can be simply solved with the form of $c_v = G_f * (-p_v - \Omega G)$ and $c_i = G_f * (-p_i - \Omega G)$. Combining the boundary condition that $c_v(\mathbf{x} \rightarrow \infty) = c_{0v}$ and $c_i(\mathbf{x} \rightarrow \infty) = c_{0i}$, we have

$$c_v(\mathbf{x}) = \frac{1}{4\pi D_v} \int_{\mathcal{N}} \frac{\mathbf{w}'_v \cdot (\mathbf{b}' \times d\boldsymbol{\ell}')}{\|\mathbf{x} - \mathbf{x}'\|} - \frac{\Omega G}{4\pi D_v} \int_{\mathcal{V}} \frac{d\mathbf{x}'}{\|\mathbf{x} - \mathbf{x}'\|} + c_{0v} \quad (C.7)$$

$$c_i(\mathbf{x}) = \frac{1}{4\pi D_i} \int_{\mathcal{N}} \frac{\mathbf{w}'_i \cdot (\mathbf{b}' \times d\boldsymbol{\ell}')}{\|\mathbf{x} - \mathbf{x}'\|} - \frac{\Omega G}{4\pi D_i} \int_{\mathcal{V}} \frac{d\mathbf{x}'}{\|\mathbf{x} - \mathbf{x}'\|} + c_{0i} \quad (C.8)$$

The total climb velocity is simply the sum (with the proper sign) of individual velocities, and is consistent with Eq. (53). Actually, the total velocity can be also obtained by solving the net flux equation (even without omitting the vacancy-interstitial recombination), which reads

$$D_v \Delta c_v - D_i \Delta c_i = -p_{climb} \quad (C.9)$$

Thus, we have

$$D_v c_v - D_i c_i = \int_{\mathcal{N}} \frac{\mathbf{w}'_c \cdot (\mathbf{b}' \times d\boldsymbol{\ell}')}{4\pi \|\mathbf{x} - \mathbf{x}'\|} + D_v c_{0v} - D_i c_{0i} \quad (C.10)$$

We now consider a case with a single prismatic dislocation loop, which is parameterized by the loop radius R and the Burgers vector $(0, 0, b)$ as $(R \cos \theta, -R \sin \theta, 0)$. The loop shrinks or grows by absorbing point defects under its self-stress. The climb force and velocity are constants around the entire loop. The concentration of vacancies and interstitials at the point $(R, 0, 0)$ reads

$$c_v(R, 0, 0) = \frac{w_v b R}{4\pi D_v} \int_0^{2\pi} \frac{d\theta}{\sqrt{(R - R \cos \theta)^2 + R \sin^2 \theta + r_d^2}} - \frac{\Omega G}{4\pi D_v} \int_V \frac{d\mathbf{x}'}{\|\mathbf{x} - \mathbf{x}'\|} + c_{0v} \quad (\text{C.11})$$

$$c_i(R, 0, 0) = \frac{w_i b R}{4\pi D_i} \int_0^{2\pi} \frac{d\theta}{\sqrt{(R - R \cos \theta)^2 + R \sin^2 \theta + r_d^2}} - \frac{\Omega G}{4\pi D_i} \int_V \frac{d\mathbf{x}'}{\|\mathbf{x} - \mathbf{x}'\|} + c_{0i} \quad (\text{C.12})$$

The first integral can be estimated by complete elliptic integrals of the first kind K and the second integral is assumed to be a constant S related with the surface of the domain. We then have

$$c_v(R, 0, 0) = \frac{w_v b}{\pi D_v} \frac{R}{\sqrt{4R^2 + r_d^2}} K \left(\frac{R}{\sqrt{4R^2 + r_d^2}} \right) - \frac{\Omega G S}{4\pi D_v} + c_{0v} \quad (\text{C.13})$$

$$c_i(R, 0, 0) = \frac{w_i b}{\pi D_i} \frac{R}{\sqrt{4R^2 + r_d^2}} K \left(\frac{R}{\sqrt{4R^2 + r_d^2}} \right) - \frac{\Omega G S}{4\pi D_i} + c_{0i} \quad (\text{C.14})$$

On the other hand, the concentration on the dislocation reaches the equilibrium condition as

$$c_v(R, 0, 0) = c_{0v} \exp(-f_{cl} \Omega / b k_B T) \quad (\text{C.15})$$

$$c_i(R, 0, 0) = c_{0i} \exp(+f_{cl} \Omega / b k_B T) \quad (\text{C.16})$$

This finally yields

$$w_v = \frac{\pi D_v}{b} \frac{\sqrt{4R^2 + r_d^2}}{RK} \left(c_{0v} \exp(-f_{cl} \Omega / b k_B T) - c_{0v} + \frac{\Omega G S}{4\pi D_v} \right) \quad (\text{C.17})$$

$$w_i = \frac{\pi D_i}{b} \frac{\sqrt{4R^2 + r_d^2}}{RK} \left(c_{0i} \exp(+f_{cl} \Omega / b k_B T) - c_{0i} + \frac{\Omega G S}{4\pi D_i} \right) \quad (\text{C.18})$$

Applying the asymptotic behavior that $K(k) \sim \ln(4/\sqrt{1-k^2})$ for $k \rightarrow 1$, $R \gg r_d$ and $f_{cl} \Omega \ll b k_B T$, we have

$$w_v = \frac{2\pi D_v}{\ln(8R/r_d)b} \left(-c_{0v} f_{cl} \Omega / b k_B T + \frac{\Omega G S}{4\pi D_v} \right) \quad (\text{C.19})$$

$$w_i = \frac{2\pi D_i}{\ln(8R/r_d)b} \left(+c_{0i} f_{cl} \Omega / b k_B T + \frac{\Omega G S}{4\pi D_i} \right) \quad (\text{C.20})$$

Similarly, the total velocity can be solved as

$$D_v(c_{0v} \exp(-f_{cl} \Omega / b k_B T) - c_{0v}) - D_i(c_{0i} \exp(+f_{cl} \Omega / b k_B T) - c_{0i}) = \frac{wb}{\pi} \frac{R}{\sqrt{4R^2 + r_d^2}} K \left(\frac{R}{\sqrt{4R^2 + r_d^2}} \right) \quad (\text{C.21})$$

Again applying the asymptotic behavior that $K(k) \sim \ln(4/\sqrt{1-k^2})$ for $k \rightarrow 1$, $R \gg r_d$, we have

$$w = \frac{2\pi}{\ln(8R/r_d)b} (D_v c_{0v} \exp(-f_{cl} \Omega / b k_B T) - D_v c_{0v} - D_i c_{0i} \exp(+f_{cl} \Omega / b k_B T) + D_i c_{0i}) \quad (\text{C.22})$$

And with $f_{cl} \Omega \ll b k_B T$, we have

$$w \approx -\frac{2\pi f_{cl} \Omega}{\ln(8R/r_d)b^2 k_B T} (D_v c_{0v} - D_i c_{0i}) = -\frac{2\pi f_{cl} \Omega}{\ln(8R/r_d)b^2 k_B T} \left(D_{v0} e^{-\frac{E_v^s + \frac{\Delta \Omega_{FE}^s}{3} \text{tr}(\sigma)}{k_B T}} - D_{i0} e^{-\frac{E_i^s - \frac{\Delta \Omega_{FI}^s}{3} \text{tr}(\sigma)}{k_B T}} \right) \quad (\text{C.23})$$

Where $E^s = E^f + E^m$. Under isotropic diffusion, and with no bias, the climb velocity of a dislocation loop does not depend on the production and interaction terms and will reduce to the thermal case.

Appendix D. Concept of different biases

Bias factors are key parameters that influence the reaction rates between mobile defects and sinks, including dislocations and grain boundaries. They are obtained by solving the diffusion problem in the vicinity of a sink through an effective medium at a stationary state. Analytical expressions of bias factors can be obtained only with simple geometries and simplified description

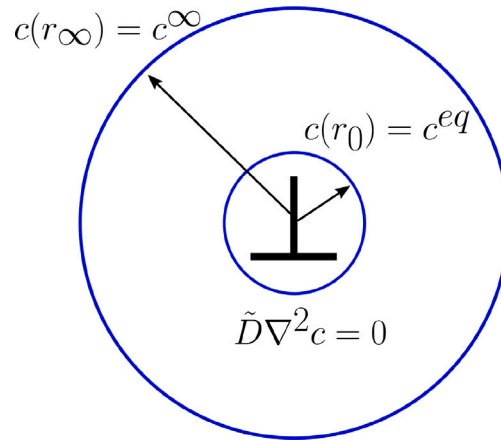


Fig. D.1. Diagram for the elastic interaction bias.

of point defect properties (Woo and Goesele, 1983; Woo, 1988). The full complexity of this problem entails dedicated numerical simulations, such as molecular dynamics (Wolfer, 2007) and kinetic Monte-Carlo (Da Fonseca et al., 2023) simulations.

It is widely accepted that in describing the reaction rate of point defects with dislocations, their elastic interaction, the diffusional anisotropy of point defects, and external stresses are the most influential bias factors. First, the elastic interaction of point defects with dislocations modifies the migration of the point defects towards sinks, and thus their flux. The vacancies and interstitials have different interaction energies with dislocations, which gives rise to the different reaction rates. This is the origin of the conventional dislocation bias. For example, the analytical dislocation bias can be solved within the simple scheme shown in Fig. D.1. Assuming that far from the sink, a reservoir of point-defect exists, and is maintained at a constant concentration c^∞ , the concentration of defects around an infinite long straight dislocation, $c(r)$, can be solved from the steady-state diffusion equation

$$\nabla \cdot \mathbf{J} = -\nabla \cdot \left[\nabla(Dc) + \frac{Dc}{k_B T} \nabla E \right] \quad (\text{D.1})$$

together with the boundary conditions, $c(r = r_\infty) = c^\infty$, and $c(r = r_0) = c^{eq}$, where $D(r)$ is the diffusion coefficient, $E(r)$ is the interaction energy between the dislocation and defects, r_0 is the dislocation core radius, and c^{eq} is the equilibrium concentration of defects on the dislocation. The ratio of the actual flux of defects towards the dislocation to that without an interaction energy, i.e. $E(r) = 0$ is defined as the dislocation bias factor. A detailed discussion of dislocation bias can be found in Wolfer (2007). Inspired by this idea, we split the flux and introduce the dislocation bias into our framework via Eq. (45). It is worth noting that this dislocation bias is an effective parameter characterizing the ‘average’ reaction rate between defects and dislocations when accounting for their elastic interactions.

Second, in the presence of the external applied stress, it is found that the reaction rates depend on the relative orientation of dislocations with respect to the external stress. This gives rise to a stress-induced preferred absorption (SIPA), often used to explain irradiation creep. In the present work, external forces are not considered; therefore, this effect is ignored. However, it can be effectively integrated into the framework by further modifying the bias factor. For instance, considering a defect as a spherical inhomogeneity, the long-range strain field of a dislocation can detect the perturbation of the applied stress field near the point defect, and then gives rise to an inhomogeneity interaction. This interaction enables point defects to distinguish more favorable dislocations to migrate towards. By assuming that the dipole tensor is the same at the stable and saddle points, Heald and Speight (1975) have shown that this effect, denoted as SIPA-I, can be characterized by a factor $Z^I(\vec{\sigma})$ as

$$Z^I(\vec{\sigma}) = Z^0 \left(1 + \frac{\Delta Z^I(\vec{\sigma})}{Z^0} \right) \quad (\text{D.2})$$

where Z^0 is the bias factor without applied stress. $\Delta Z^I(\vec{\sigma})$ is the deviation due to the inhomogeneity interaction, depending on the defect, the stress, and the dislocation direction. However, an applied stress can also influence the jump barrier of point defects as a function of the jump direction. This is the so-called elastodiffusion (SIPA-AD), and has been thoroughly studied by Woo and Goesele (1983) and Borodin and Ryazanov (1994), i.e.,

$$Z^{AD}(\vec{\sigma}) = Z^0 \left(1 + \frac{\Delta Z^{AD}(\vec{\sigma})}{Z^0} \right) \quad (\text{D.3})$$

where $\Delta Z^{AD}(\vec{\sigma})$ describes the effects of the saddle point anisotropy due to the stress. Contrary to the SIPA-I model, defects have different properties at stable and saddle positions in this model.

In addition, the crystallographic anisotropy of non-cubic metals makes the diffusion rates of point defects dependent on the crystallographic directions, even in the absence of the external applied stress. This results in a difference in the absorption rates at dislocations, which is known as diffusional anisotropy difference (DAD). It should be pointed out that this DAD in non-cubic crystal

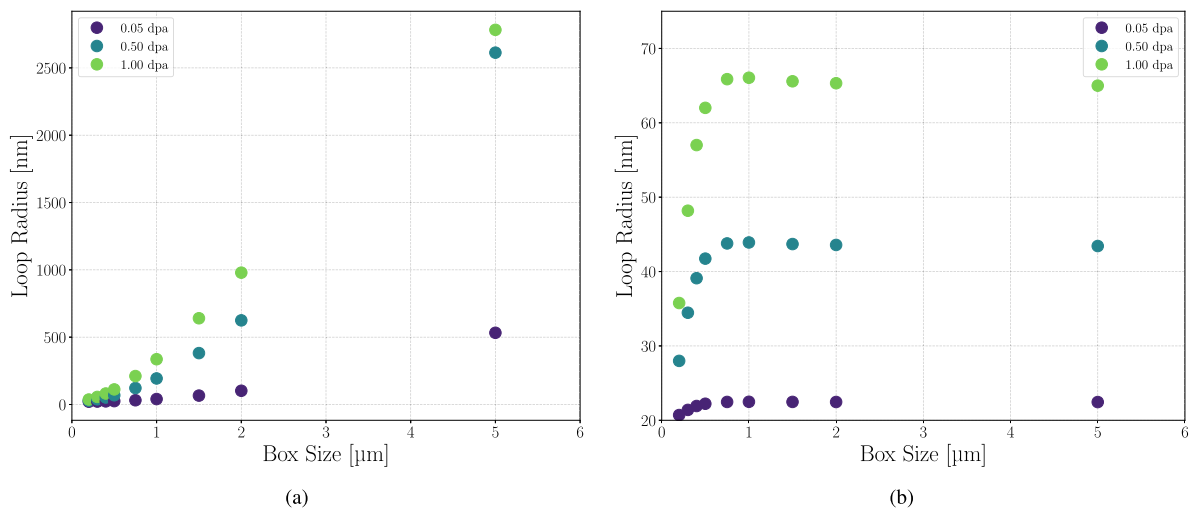


Fig. E.2. Effect of simulation domain size on dislocation loop growth. A 20 nm basal vacancy loop is placed at the center of a cubic domain with side length L varying from 0.2 to 5.0 μm , where the volume is L^3 . Both cases use $G_0 = 10^{-7}$ dpa/s, $p^m = 0.7$ for interstitials while vacancies diffuse isotropically. Marker colors correspond to dose levels of 0.05, 0.50, and 1.00 dpa. (a) No recombination of defects: the loop radius increases significantly with box size due to classical diffusion-limited behavior. (b) With recombination of defects: mobile defect recombination flattens the concentration profile and suppresses boundary effects, leading to convergence in loop radius for box sizes $L > 0.8 \mu\text{m}$.

is a zero-order effect associated with the intrinsic diffusional anisotropy, and is distinguished from the elastodiffusion. This DAD effect is explicitly included within our model, and the analytical bias factor associated with DAD can be obtained by investigating the reaction of a single dislocation and anisotropically diffusing defects (Woo, 1988). Lastly, many experiments and atomic simulations have shown that under neutron irradiation, a large fraction of the defects form clusters (Guerard et al., 1980; Bacon et al., 1999). The self-interstitial atom (SIA) clusters migrate one- or two-dimensionally, which results in the biased reaction rate for different types of sinks. Smaller interstitial clusters have generally been found to be mobile and diffuse highly anisotropically, contributing to mass transport to the $\langle a \rangle$ axes of the hcp crystal, meanwhile large interstitial clusters have been found to collapse into platelets forming dislocation loops. These findings lead to the development of the production bias model (PBM), which succeeds in studying many radiation-induced phenomena, e.g., the void swelling, the void lattice formation in cubic crystals (Golubov et al., 2012), and the radiation growth in hexagonal crystals (Barashev et al., 2015).

Appendix E. Effect of simulation volume

As boundary conditions and crystal size are important influences on loop kinetics, we assess here the effect of simulation volume on loop evolution by performing a series of simulations varying the box side length from $L = 0.2 \mu\text{m}$ to $L = 5 \mu\text{m}$ as shown in Fig. E.2. A single vacancy dislocation loop of 20 nm radius with Burgers vector of $\mathbf{b} = c/2[0001]$ is placed at the center of a cubic simulation domain. Only single interstitials and vacancies are considered following Table 1. The defect production rate is $G_0 = 10^{-7}$ dpa/s, and anisotropic diffusion is applied to single interstitials, using migration energies of $E_i^{m(33)} = 0.158$ eV and $E_i^{m(11)} = E_i^{m(22)} = 0.056$ eV, corresponding to an anisotropy factor of $p^m = 0.7$. Vacancies are treated as an isotropic diffuser with $E_v^m = 0.9$ eV. Dirichlet boundary conditions are imposed at the surfaces, where the defect concentrations are set equal to their thermal equilibrium values and corrected by the stress fields imposed by the dislocations. We compare two cases: (1) without defect recombination, shown in Fig. E.2(a), and (2) with defect recombination, shown in Fig. E.2(b). For each simulation, we extract the loop radius at doses of 0.05, 0.50, and 1.00 dpa. The results of these studies reveal that the recombination of mobile defects plays a critical role and minimizes boundary effects at the crystal center. When recombination is active, the concentration profiles flatten near the center of the crystal, and loop growth in this region becomes relatively insensitive to crystal size above a critical value. In contrast, when recombination is disabled, the concentration profile retains its classical parabolic form, and the net flux to the loop scales with L^2 , leading to a strong dependence of loop growth on system size and resulting in unrealistically large loop growth rates in larger boxes. These results establish that, for the production rate, diffusion coefficients, and reaction rates used, finite-size effects are effectively eliminated for domain sizes $L \gtrsim 0.8 \mu\text{m}$. Therefore, to eliminate boundary effects when studying individual loop kinetics, we use a crystal with side length greater than 1 μm for this work. These findings further emphasize the necessity of including recombination processes in spatially resolved models to accurately capture finite-size effects, and further show a converging behavior above a critical size for individual dislocations centered in a crystal or in a bulk environment.

Data availability

Data will be made available on request.

References

- Adamson, R.B., Coleman, C.E., Griffiths, M., 2019. Irradiation creep and growth of zirconium alloys: A critical review. *J. Nucl. Mater.* 521, 167–244.
- Ayas, C., Van Dommelen, J., Deshpande, V., 2014. Climb-enabled discrete dislocation plasticity. *J. Mech. Phys. Solids* 62, 113–136.
- Bacon, D., Gao, F., Osetsky, Y.N., 1999. Computer simulation of displacement cascades and the defects they generate in metals. *Nucl. Instrum. Methods Phys. Res. Sect. B: Beam Interactions Mater. Atoms* 153 (1–4), 87–98.
- Bacon, D.J., Osetsky, Y.N., Stoller, R., Voskoboynikov, R.E., 2003. MD description of damage production in displacement cascades in copper and α -iron. *J. Nucl. Mater.* 323 (2–3), 152–162.
- Bakó, B., Clouet, E., Dupuy, L.M., Blétry, M., 2011. Dislocation dynamics simulations with climb: kinetics of dislocation loop coarsening controlled by bulk diffusion. *Phil. Mag.* 91 (23), 3173–3191.
- Balluffi, R.W., Seidman, D.N., 1965. Diffusion-limited climb rate of a dislocation: Effect of climb motion on climb rate. *J. Appl. Phys.* 36 (9), 2708–2711.
- Barashev, A.V., Golubov, S.I., Stoller, R.E., 2015. Theoretical investigation of microstructure evolution and deformation of Zirconium under neutron irradiation. *J. Nucl. Mater.* 461, 85–94.
- Bardeen, J., Herring, C., 1952. Diffusion in alloys and the Kirkendall effect. In: *Imperfections in Nearly Perfect Crystals*. Wiley, pp. 261–288.
- Bilby, B.A., Gardner, L., Stroh, A.N., 1957. Continuous distributions of dislocations and the theory of plasticity. In: *Proceedings of the Ninth International Congress on Applied Mechanics*. Bruxelles, pp. 35–44.
- Borodin, V., Ryazanov, A., 1994. The effect of diffusion anisotropy on dislocation bias and irradiation creep in cubic lattice materials. *J. Nucl. Mater.* 210 (3), 258–267.
- Brailsford, A., Bullough, R., 1972. The rate theory of swelling due to void growth in irradiated metals. *J. Nucl. Mater.* 44 (2), 121–135.
- Breidi, A., Dudarev, S., 2022. Dislocation dynamics simulation of thermal annealing of a dislocation loop microstructure. *J. Nucl. Mater.* 562, 153552.
- Buckley, S., Bullough, R., Hayns, M., 1980. The direct observation of irradiation damage in zirconium and its alloys. *J. Nucl. Mater.* 89 (2–3), 283–295.
- Cai, W., Arsenlis, A., Weinberger, C.R., Bulatov, V.V., 2006. A non-singular continuum theory of dislocations. *J. Mech. Phys. Solids* 54 (3), 561–587.
- Carpenter, G., Murgatroyd, R., Rogerson, A., Watters, J., 1981. Irradiation growth of zirconium single crystals. *J. Nucl. Mater.* 101 (1–2), 28–37.
- Carpenter, G., Zee, R., Rogerson, A., 1988. Irradiation growth of zirconium single crystals: A review. *J. Nucl. Mater.* 159, 86–100.
- Choi, S.I., Lee, G.-G., Kwon, J., Kim, J.H., 2016. Modeling of sink-induced irradiation growth of single-crystal and polycrystal zirconiums in nuclear reactors. *J. Nucl. Mater.* 468, 56–70.
- Christiaen, B., Domain, C., Thuinet, L., Ambard, A., Legris, A., 2020. Influence of vacancy diffusional anisotropy: Understanding the growth of zirconium alloys under irradiation and their microstructure evolution. *Acta Mater.* 195, 631–644.
- Christien, F., Barbu, A., 2009. Cluster dynamics modelling of irradiation growth of zirconium single crystals. *J. Nucl. Mater.* 393 (1), 153–161.
- Cui, W., Cui, Y., Liu, W., 2022. New insights into spatio-temporal dynamics of irradiation defects rafting. *J. Nucl. Mater.* 568, 153840.
- Da Fonseca, D., Onimus, F., Momprou, F., Marinica, M.-C., de Sonis, E., Clouet, E., Jourdan, T., 2023. Numerical investigation of dislocation climb under stress and irradiation. *Acta Mater.* 242, 118431.
- Dai, C., Balogh, L., Yao, Z., Daymond, M.R., 2017. The habit plane of $\langle a \rangle$ -type dislocation loops in α -zirconium: an atomistic study. *Phil. Mag.* 97 (12), 944–956.
- Dai, Y., Li, J., Liu, B., 2009. Long-range empirical potential model: Extension to hexagonal close-packed metals. *J. Phys.: Condens. Matter* 21 (38), 385402.
- Dai, C., Saidi, P., Topping, M., Béland, L., Yao, Z., Daymond, M., 2019. A mechanism for basal vacancy loop formation in zirconium. *Scr. Mater.* 172, 72–76.
- Danas, K., Deshpande, V., 2013. Plane-strain discrete dislocation plasticity with climb-assisted glide motion of dislocations. *Modelling Simul. Mater. Sci. Eng.* 21 (4), 045008.
- De Diego, N., Osetsky, Y.N., Bacon, D., 2002. Mobility of interstitial clusters in alpha-zirconium. *Met. Mater. Trans. A* 33, 783–789.
- Dederichs, P., Schroeder, K., 1978. Anisotropic diffusion in stress fields. *Phys. Rev. B* 17 (6), 2524.
- Derlet, P., Dudarev, S., 2020. Microscopic structure of a heavily irradiated material. *Phys. Rev. Mater.* 4 (2), 023605.
- Dudarev, S., Ma, P.-W., 2018. Elastic fields, dipole tensors, and interaction between self-interstitial atom defects in bcc transition metals. *Phys. Rev. Mater.* 2 (3), 033602.
- Dymant, F., Libanati, C., 1968. Self-diffusion of Ti, Zr, and Hf in their hcp phases, and diffusion of Nb 95 in hcp Zr. *J. Mater. Sci.* 3, 349–359.
- Freyssoldt, C., Grabowski, B., Hickel, T., Neugebauer, J., Kresse, G., Janotti, A., Van de Walle, C.G., 2014. First-principles calculations for point defects in solids. *Rev. Modern Phys.* 86 (1), 253–305.
- Fu, B., Fitzgerald, S., Hou, Q., Wang, J., Li, M., 2017. Effect of collision cascades on dislocations in tungsten: A molecular dynamics study. *Nucl. Instrum. Methods Phys. Res. Sect. B: Beam Interactions Mater. Atoms* 393, 169–173.
- Gao, S., Ali, M.A., Hartmaier, A., 2019. Influence of rafted microstructures on creep in Ni-base single crystal superalloys: a 3D discrete dislocation dynamics study. *Modelling Simul. Mater. Sci. Eng.* 28 (2), 025001.
- Geslin, P.-A., Appolaire, B., Finel, A., 2014. A phase field model for dislocation climb. *Appl. Phys. Lett.* 104 (1), 011903.
- Ghorbani, A., Luo, Y., Saidi, P., Beland, L.K., 2024. Anisotropic diffusion of radiation-induced self-interstitial clusters in HCP zirconium: A molecular dynamics and rate-theory assessment. *Scr. Mater.* 238, 115755.
- Van der Giessen, E., Needleman, A., 1995. Discrete dislocation plasticity: a simple planar model. *Modelling Simul. Mater. Sci. Eng.* 3 (5), 689.
- Gilbert, R., Farrell, K., Coleman, C., 1979. Damage structure in zirconium alloys neutron irradiated at 573 to 923 K. *J. Nucl. Mater.* 84 (1–2), 137–148.
- Golubov, S., Barashev, A., Stoller, R., 2012. 1.13-Radiation damage theory. *Compr. Nucl. Mater.* 1, 357.
- Goyal, A., Mathew, K., Hennig, R.G., Chernatynskiy, A., Stanek, C.R., Murphy, S.T., Andersson, D.A., Phillpot, S.R., Uberuaga, B.P., 2019. The conundrum of relaxation volumes in first-principles calculations of charged defects in UO₂. *Appl. Sci.* 9 (24), 5276.
- Greenwood, G., Foreman, A., Rimmer, D., 1959. The role of vacancies and dislocations in the nucleation and growth of gas bubbles in irradiated fissile material. *J. Nucl. Mater.* 1 (4), 305–324.
- Griffiths, M., 1988. A review of microstructure evolution in zirconium alloys during irradiation. *J. Nucl. Mater.* 159, 190–218.
- Gu, Y., Xiang, Y., Quek, S.S., Srolovitz, D.J., 2015. Three-dimensional formulation of dislocation climb. *J. Mech. Phys. Solids* 83, 319–337.
- Guerard, B.v., Grasse, D., Peisl, J., 1980. Structure of defect cascades in fast-neutron-irradiated aluminum by diffuse X-ray scattering. *Phys. Rev. Lett.* 44 (4), 262.
- Heald, P., Speight, M., 1975. Point defect behaviour in irradiated materials. *Acta Metall.* 23 (11), 1389–1399.
- Hirth, J.P., Lothe, J., Mura, T., 1983. Theory of dislocations.
- Hood, G., 1986. Diffusion and vacancy properties of α -Zr. *J. Nucl. Mater.* 139 (3), 179–184.
- Hood, G., 1988. Point defect diffusion in α -Zr. *J. Nucl. Mater.* 159, 149–175.
- Horvath, J., Dymant, F., Mehrer, H., 1984. Anomalous self-diffusion in a single crystal of α -zirconium. *J. Nucl. Mater.* 126 (3), 206–214.
- Hulse, R., Race, C.P., 2021. An atomistic modelling study of the properties of dislocation loops in zirconium. *J. Nucl. Mater.* 546, 152752.
- Ji, C., Li, Y., Sun, Z., Cui, A., Xin, Y., Cui, Y., 2024. Multiscale-informed irradiation growth model of Zr-Sn-Nb alloys. *Int. J. Plast.* 540, 104177.
- Jiang, S., Rachh, M., Xiang, Y., 2017. An efficient high order method for dislocation climb in two dimensions. *Multiscale Model. Simul.* 15 (1), 235–253.
- Jostons, A., Kelly, P., Blake, R., 1977. The nature of dislocation loops in neutron irradiated zirconium. *J. Nucl. Mater.* 66 (3), 236–256.
- Kelly, P., Blake, R., Jostons, A., 1976. An interpretation of corduroy contrast in neutron irradiated zirconium. *J. Nucl. Mater.* 59 (3), 307–315.
- Kim, Y.-M., Lee, B.-J., Baskes, M., 2006. Modified embedded-atom method interatomic potentials for Ti and Zr. *Phys. Rev. B* 74 (1), 014101.

- Kim, Y.-S., et al., 2018. Molecular dynamics simulations of the coupled effects of strain and temperature on displacement cascades in α -zirconium. *Nucl. Eng. Technol.* 50 (6), 907–914.
- Kobylansky, G., Novoselov, A., Ostrovsky, Z., Obukhov, A., Shishin, V., Shishov, V., Nikulina, A., Perehud, M., Mahmood, S., White, D., et al., 2008. Irradiation-induced growth and microstructure of recrystallized, cold worked and quenched Zircaloy-2, NSF, and E635 alloys. *J. ASTM Int.* 5 (4), 1–19.
- Kohnert, A.A., Capolungo, L., 2022. The kinetics of static recovery by dislocation climb. *Npj Comput. Mater.* 8 (1), 104.
- Kröner, E., 1960. Allgemeine Kontinuumsmechanik der Versetzungen Und Eigenspannungen. *Arch. Ration. Mech. Anal.* 4 (4), 273–334.
- Larché, F., Cahn, J., 1982. The effect of self-stress on diffusion in solids. *Acta Metall.* 30 (10), 1835–1845.
- Le Bacq, O., Willaime, F., Pasturel, A., 1999. Unrelaxed vacancy formation energies in group-IV elements calculated by the full-potential linear muffin-tin orbital method: Invariance with crystal structure. *Phys. Rev. B* 59 (13), 8508.
- Lee, E., 1969. Elastic-Plastic Deformation at Finite Strains. *J. Appl. Mech.* 36 (1), 1–6.
- Li, Y., Ghoniem, N., 2020. Cluster dynamics modeling of irradiation growth in single crystal Zr. *J. Nucl. Mater.* 540, 152312.
- Li, S., Li, Y., Cao, Z., Ding, Y., Wu, X., Zhang, R., Ran, G., 2024a. Abnormal growth of loops by repulsive interaction between loops and dislocations. *J. Nucl. Mater.* 605, 155589.
- Li, S., Li, Y., Cao, Z., Ding, Y., Wu, X., Zhang, R., Ran, G., 2025. Abnormal growth of loops by repulsive interaction between loops and dislocations. *J. Nucl. Mater.* 605, 155589.
- Li, Y., Po, G., Cui, Y., Ghoniem, N., 2023. Prismatic-to-basal plastic slip transition in zirconium. *Acta Mater.* 242, 118451.
- Li, Y., Po, G., Ghoniem, N., 2020. Coupled cluster-dislocation dynamics of irradiation-induced defects. *Materialia* 14, 100891.
- Li, S., Xie, H., Yu, L., Xiao, X., 2024b. A steady-state irradiation creep and thermal creep model for zirconium alloys. *Int. J. Plast.* 172, 103814.
- Liu, S.-M., Beyerlein, I.J., Han, W.-Z., 2020. Two-dimensional vacancy platelets as precursors for basal dislocation loops in hexagonal zirconium. *Nat. Commun.* 11 (1), 5766.
- Liu, S.-M., Han, W.-Z., 2021. Transmission electron microscopy characterization of dislocation loops in irradiated zirconium. *Tungsten* 3 (4), 470–481.
- Liu, B., Raabe, D., Roters, F., Arsenlis, A., 2014. Interfacial dislocation motion and interactions in single-crystal superalloys. *Acta Mater.* 79, 216–233.
- Long, F., Roy, R., Daymond, M., 2024. Electron microscopy characterization of proton irradiation induced growth in pure Zr. *J. Nucl. Mater.* 599, 155243.
- Lothe, J., 1960. Theory of dislocation climb in metals. *J. Appl. Phys.* 31 (6), 1077–1087.
- Malvern, L.E., 1977. Introduction to the Mechanics of a Continuous Medium, first ed. Prentice Hall.
- March-Rico, J.F., Blondel, S., Wirth, B.D., 2025. Atomistically-informed cluster dynamics modeling of concurrent interstitial and vacancy a-loop evolution in irradiated alpha-zirconium. *Acta Mater.* 120777.
- March-Rico, J.F., Wirth, B.D., 2022. Defect cluster configurations and mobilities in α -Zr: A comparison of the BMD19 and M07 interatomic potentials. *J. Nucl. Mater.* 559, 153441.
- Maron, M., Li, Y., Lalani, I., Baker, K., Flores, B.R., Black, T., Hollenbeck, J., Ghoniem, N., Po, G., 2024. Spatially-resolved cluster dynamics modeling of irradiation growth. *Int. J. Plast.* 177, 103989.
- Mendelev, M.I., Ackland, G.J., 2007. Development of an interatomic potential for the simulation of phase transformations in zirconium. *Phil. Mag. Lett.* 87 (5), 349–359.
- Mikhlin, A., Osetsky, Y.N., Kapinos, V., 1994. On the anisotropic migration of point defects in hcp zirconium. *Phil. Mag. A* 70 (1), 25–33.
- MoDeLib2, 2025. MoDeLib2. <https://github.com/{MoDeLib2}/{MoDeLib2}>.
- Monnet, G., Devincere, B., Kubin, L., 2004. Dislocation study of prismatic slip systems and their interactions in hexagonal close packed metals: application to zirconium. *Acta Mater.* 52 (14), 4317–4328.
- Mordehai, D., Clouet, E., Fivel, M., Verdier, M., 2008. Introducing dislocation climb by bulk diffusion in discrete dislocation dynamics. *Phil. Mag.* 88 (6), 899–925.
- Northwood, D., Gilbert, R., Bahen, L., Kelly, P., Blake, R., Jostons, A., Madden, P., Faulkner, D., Bell, W., Adamson, R., 1979. Characterization of neutron irradiation damage in zirconium alloys—an international “round-robin” experiment. *J. Nucl. Mater.* 79 (2), 379–394.
- Onsager, L., 1931. Reciprocal Relations in Irreversible Processes. I. *Phys. Rev.* 37, 405–426.
- Osetsky, Y.N., Bacon, D., De Diego, N., 2002. Anisotropy of point defect diffusion in alpha-zirconium. *Met. Mater. Trans. A* 33 (3), 777–782.
- Ouyang, Y., Wu, J., Zheng, M., Chen, H., Tao, X., Du, Y., Peng, Q., 2018. An interatomic potential for simulation of defects and phase change of zirconium. *Comput. Mater. Sci.* 147, 7–17.
- Peach, M., Koehler, J., 1950. The Forces Exerted on Dislocations and the Stress Fields Produced by Them. *Phys. Rev.* 80 (3), 436–439.
- Peng, Q., Ji, W., Huang, H., De, S., 2012. Stability of self-interstitial atoms in hcp-Zr. *J. Nucl. Mater.* 429 (1–3), 233–236.
- Perez, R.A., Nakajima, H., Dymant, F., 2003. Diffusion in α -Ti and Zr. *Mater. Trans.* 44 (1), 2–13.
- Po, G., Ghoniem, N., 2014. A variational formulation of constrained dislocation dynamics coupled with heat and vacancy diffusion. *J. Mech. Phys. Solids* 66, 103–116.
- Po, G., Huang, Y., Li, Y., Baker, K., Flores, B.R., Black, T., Hollenbeck, J., Ghoniem, N., 2022. A model of thermal creep and annealing in finite domains based on coupled dislocation climb and vacancy diffusion. *J. Mech. Phys. Solids* 169, 105066.
- Rouchette, H., Thuinet, L., Legris, A., Ambard, A., Domain, C., 2014. Influence of shape anisotropy of self-interstitials on dislocation sink efficiencies in Zr: Multiscale modeling. *Phys. Rev. B* 90 (1), 014104.
- Rovelli, I., Dudarev, S., Sutton, A., 2018. Statistical model for diffusion-mediated recovery of dislocation and point-defect microstructure. *Phys. Rev. E* 98 (4), 043002.
- Ruiz, D.H., Gribaudo, L.M., Monti, A.M., 2005. Self-diffusion in the hexagonal structure of Zirconium and Hafnium: computer simulation studies. *Mater. Res.* 8, 431–434.
- Rungamornrat, J., Senjuntichai, T., 2009. Regularized boundary integral representations for dislocations and cracks in smart media. *Smart Mater. Struct.* 18 (7), 074010.
- Saidi, P., Wang, Z., Mao, Y., Béland, L.K., Yao, Z., Daymond, M.R., 2020. A method for calculation of bias factor in anisotropic mediums, application to α -zirconium. *J. Nucl. Mater.* 528, 151882.
- Sakaël, C., Domain, C., Ambard, A., Thuinet, L., Legris, A., 2023. Modelling of zirconium growth under irradiation and annealing conditions. *Int. J. Plast.* 168, 103699.
- Sakaël, C., Domain, C., Ambard, A., Thuinet, L., Legris, A., 2025. Using ab initio molecular dynamics to determine the mobility of single, di and tri self-interstitial configurations in hcp Zr. *Scr. Mater.* 255, 116386.
- Thompson, D., Tarleton, E., Roberts, S., Fitzgerald, S., 2018. Interstitial-mediated dislocation climb and the weakening of particle-reinforced alloys under irradiation. *Phys. Rev. Mater.* 2 (8), 080601.
- Tomé, C., Cecatto, H., Savino, E., 1982. Point-defect diffusion in a strained crystal. *Phys. Rev. B* 25 (12), 7428.
- Tournadre, L., Onimus, F., Béchade, J.-L., Gilbon, D., Cloué, J.-M., Mardon, J.-P., Feaugas, X., Toader, O., Bachelet, C., 2012. Experimental study of the nucleation and growth of c-component loops under charged particle irradiations of recrystallized Zircaloy-4. *J. Nucl. Mater.* 425 (1–3), 76–82.
- Varvenne, C., Bruneval, F., Marinica, M.-C., Clouet, E., 2013. Point defect modeling in materials: Coupling ab initio and elasticity approaches. *Phys. Rev. B—Condensed Matter Materials Phys.* 88 (13), 134102.
- Varvenne, C., Mackain, O., Clouet, E., 2014. Vacancy clustering in zirconium: An atomic-scale study. *Acta Mater.* 78, 65–77.

- Vérité, G., Domain, C., Fu, C.-C., Gasca, P., Legris, A., Willaime, F., 2013. Self-interstitial defects in hexagonal close packed metals revisited: Evidence for low-symmetry configurations in Ti, Zr, and Hf. *Phys. Rev. B—Condensed Matter Materials Phys.* 87 (13), 134108.
- Voskoboinikov, R., Osetsky, Y.N., Bacon, D., 2006. Identification and morphology of point defect clusters created in displacement cascades in α -zirconium. *Nucl. Instrum. Methods Phys. Res. Sect. B: Beam Interactions Mater. Atoms* 242 (1–2), 530–533.
- Warwick, A.R., Boleininger, M., Dudarev, S.L., 2021. Microstructural complexity and dimensional changes in heavily irradiated zirconium. *Phys. Rev. Mater.* 5 (11), 113604.
- Warwick, A.R., Thomas, R., Boleininger, M., Koç, Ö., Zilahi, G., Ribárik, G., Hegedues, Z., Lienert, U., Ungar, T., Race, C., et al., 2023. Dislocation density transients and saturation in irradiated zirconium. *Int. J. Plast.* 164, 103590.
- Was, G., 2007. *Fundamentals of Radiation Materials Science: Metals and Alloys*. Springer.
- Wolfer, W., 1980. Correlation of radiation creep theory with experimental evidence. *J. Nucl. Mater.* 90 (1–3), 175–192.
- Wolfer, W., 2007. The dislocation bias. *J. Computer-Aided Mater. Des.* 14, 403–417.
- Woo, C., 1984. Irradiation creep due to elastodiffusion. *J. Nucl. Mater.* 120 (1), 55–64.
- Woo, C., 1988. Theory of irradiation deformation in non-cubic metals: effects of anisotropic diffusion. *J. Nucl. Mater.* 159, 237–256.
- Woo, C.H., 2005. Modeling irradiation damage accumulation in crystals. In: *Handbook of Materials Modeling: Methods*. Springer, pp. 959–986.
- Woo, C., Goesele, U., 1983. Dislocation bias in an anisotropic diffusive medium and irradiation growth. *J. Nucl. Mater.* 119 (2–3), 219–228.
- Woo, C., Liu, X., 2007. Atomistic calculation of point-defect diffusion anisotropy and irradiation growth in α -zirconium. *Phil. Mag.* 87 (16), 2355–2369.
- Woo, C., Singh, B., 1992. Production bias due to clustering of point defects in irradiation-induced cascades. *Phil. Mag. A* 65 (4), 889–912.
- Wooding, S., Howe, L., Gao, F., Calder, A., Bacon, D., 1998. A molecular dynamics study of high-energy displacement cascades in α -zirconium. *J. Nucl. Mater.* 254 (2–3), 191–204.
- Zhang, J., Ding, S., Duan, H., 2023. Modeling of irradiation growth of Zr single crystals with diffusion anisotropy. *J. Nucl. Mater.* 577, 154295.
- Zhou, W., Tian, J., Feng, Q., Zheng, J., Liu, X., Xue, J., Qian, D., Peng, S., 2018. Molecular dynamics simulations of high-energy displacement cascades in hcp-Zr. *J. Nucl. Mater.* 508, 540–545.
- Ziegler, H., 1958. An attempt to generalize Onsager's principle, and its significance for rheological problems. *J. Appl. Math. Phys. (ZAMP)* 9 (5–6), 748–763.
- Ziegler, H., 1963. Some extremum principles in irreversible thermodynamics with application to continuum mechanics. In: Sneddon, I.N., Hill, R. (Eds.), *Progress in Solid Mechanics*. North Holland, Amsterdam, pp. 91–193.